Strömungslehre II

Formelsammlung

Ein Open-Source-Projekt für Studierende der Universität Stuttgart.

Der Quellcode dieser Formelsammlung steht auf GitHub zur Verfügung. Bei Anmerkungen, Fragen, Kritik und dem Drang, mitzuwirken, kontaktiere gerne den Herausgeber des verlinkten Repositorys. Für den Studiengang: B.Sc. Luft- und Raumfahrttechnik. Made with LaTeX and love.

Terminologie

Begriff	Bedeutung	Mathematisch
Stromlinie	Linie, die zu jedem Zeitpunkt tangential zur Geschwindigkeit verläuft (siehe Strömunglehre I)	$\Psi = \text{konstant}$
Wirbellinie	Laufen zu einer bestimmten Zeit an jedem Ort tangential zum Wirbel- oder Drehvektor $\underline{\omega}$ (analog zur Stromlinie).	-
Wirbelfaden	Gesamtheit der Wirbellinien, die durch eine Fläche A_1 ein- und eine Fläche A_2 austreten.	-
Wirbelröhre	Wirbellinien durch die Punkte einer geschlossenen Kurve, bilden die Mantelfläche der Wirbelröhre.	-
Wirbelstrom	Analog zum Volumenstrom.	$\underline{\omega} \cdot \underline{A} = \text{konstant}$
Isobare	Linie gleicher statischer Drücke in der Strömung.	$p = \text{konstant}$
Isotache	Linie gleicher Geschwindigkeit in der Strömung.	$\lvert\underline{v}\rvert = \text{konstant}$
Isokline	Linie gleicher Stromlinienneigung in der Strömung.	$\tan(\alpha) = \dfrac{v}{u} = \text{konstant}$

Symbolverzeichnis

Symbol	SI-Einheit	Bedeutung
c_p	$[-]$	Druckbeiwert
F	$[-]$	Komplexe Stromfunktion
h	$\left[\frac{\mathrm{m}}{\mathrm{s}^2}\right]$	Spezifische Enthalpie
s	$\left[\frac{\mathrm{m}}{\mathrm{s}^2 \cdot \mathrm{K}}\right]$	Spezifische Entropie
u, v, w	$\left[\frac{\mathrm{m}}{\mathrm{s}}\right]$	Geschwindigkeiten in x-, y- und z-Richtung
U_∞	$\left[\frac{\mathrm{m}}{\mathrm{s}}\right]$	Geschwindigkeit in x-Richtung, unendlich entfernt
Γ	$\left[\frac{\mathrm{m}^2}{\mathrm{s}}\right]$	Zirkulation
ϑ	$[-]$	Umlenkwinkel (Bezogen auf Anströmrichtung)
κ	$[-]$	Isentropenexponent
μ	$[-]$	Mach'scher Winkel
σ	$[-]$	Stoßwinkel (Bezogen auf Anströmrichtung)
Φ / ϕ	$\left[\frac{\mathrm{m}^2}{\mathrm{s}}\right]$	Potentialfunktion / Strömungspotential
Ψ / ψ	$\left[\frac{\mathrm{m}^2}{\mathrm{s}}\right]$	Stromlinienfunktion
Ω	$\left[\frac{1}{\mathrm{s}}\right]$	Rotation
ω	$\left[\frac{1}{\mathrm{s}}\right]$	Wirbelstärke
$\nabla(\cdots)$		Nabla-Operator (Mathematisch)
$\Delta(\cdots)$		Gradient (Mathematisch)
$\mathrm{rot}(\cdots)$		Rotation (Mathematisch)
$\mathrm{div}(\cdots)$		Divergenz (Mathematisch)

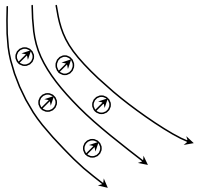
Der Übersicht wegen sind hier nur die etwas selteneren Symbole aufgeführt.
Es kann zu doppelter Symbolbelegung kommen.

Wirbeltheorie

Drehungsfrei

$\mathrm{rot}(\underline{v}_1) = 0 \;\mid\; \mathrm{div}(\rho\,\underline{v}_1) \neq 0 \;\mid\; \underline{v}_1 = \nabla(\phi)$

ϕ = Skalares Geschwindigkeitspotential



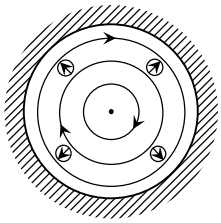
Volumenstrom

$$Q = \underline{v} \cdot \underline{A}$$

Drehungsbehaftet

$\mathrm{rot}(\underline{v}_2) \neq 0 \;\mid\; \mathrm{div}(\rho\,\underline{v}_2) = 0 \;\mid\; \rho\,\underline{v}_2 = \rho_\mathrm{b}\,\mathrm{rot}(\psi)$

ψ = Vektorielles Wirbelpotential



Wirbelstrom

$$\underline{\omega} \cdot \underline{A}$$

Wirbelstärke

$$\underline{\omega} = \frac{1}{2}\,\mathrm{rot}(\underline{v})$$

Zirkulation (positiv: ↺)

$$\Gamma = \oint_S \underline{v} \cdot \mathrm{d}\underline{s} = \oint_S v\cos(\alpha)\mathrm{d}s \quad (7.20a)$$

$$= \oiint_A \mathrm{rot}(\underline{v}) \cdot \underline{n}\mathrm{d}A = \oiint_A \underline{\Omega} \cdot \underline{n}\mathrm{d}A$$

Laplace-Operator

$$\Delta(\underline{\psi}) = \begin{pmatrix} \frac{\partial^2 \psi_x}{\partial x^2} + \frac{\partial^2 \psi_x}{\partial y^2} + \frac{\partial^2 \psi_x}{\partial z^2} \\ \frac{\partial^2 \psi_y}{\partial x^2} + \frac{\partial^2 \psi_y}{\partial y^2} + \frac{\partial^2 \psi_y}{\partial z^2} \\ \frac{\partial^2 \psi_z}{\partial x^2} + \frac{\partial^2 \psi_z}{\partial y^2} + \frac{\partial^2 \psi_z}{\partial z^2} \end{pmatrix}$$

Rotation

Allgemein

$$\underline{\Omega} = 2\,\underline{\omega} = \mathrm{rot}(\underline{v})$$

(7.11) ↩

Dichtebeständig

Allgemein

$$\underline{\Omega} = \nabla(\mathrm{div}(\underline{\psi})) - \Delta(\underline{\psi})$$

Ebene Platte

$$\Omega_z = -\Delta(\psi_z)$$

Kartesisch

$$\underline{\Omega} = \begin{pmatrix} \frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \\ \frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \\ \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \end{pmatrix}$$

Zylindrisch

$$\underline{\Omega} = \begin{pmatrix} \frac{1}{r}\frac{\partial v_z}{\partial \varphi} - \frac{\partial v_\varphi}{\partial z} \\ \frac{\partial v_r}{\partial z} - \frac{\partial v_z}{\partial r} \\ \frac{1}{r}\left(\frac{\partial}{\partial r}(r \cdot v_\varphi) - \frac{\partial v_r}{\partial \varphi}\right) \end{pmatrix}$$

Wirbelsätze

1. Helmholtz'scher Wirbelsatz

$$\mathrm{div}(\underline{\omega}) = \mathrm{div}(\underline{\Omega}) = 0$$

2. Helmholtz'scher Wirbelsatz

Reibungsfrei & Barotrop

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\frac{\underline{\omega}}{\rho}\right) = \frac{\underline{\omega}}{\rho} \cdot \nabla(\underline{v})$$

2D / Rotationssym. & Inkompr.

$$\frac{\mathrm{D}\omega}{\mathrm{D}t} = \frac{\partial \omega}{\partial t} + \underline{v} \cdot \nabla(\omega) = 0$$

Wirbeltransportgleichung

Allgemein

$$\frac{\mathrm{D}}{\mathrm{D}t}(\underline{\omega}) = \underline{\omega} \cdot \nabla(\underline{v}) + \nu\Delta(\underline{\omega})$$

2D / Rotationssym.

$$\frac{\mathrm{D}\omega}{\mathrm{D}t} = \frac{\partial \omega}{\partial t} + \underline{v} \cdot \nabla(\omega) = 0$$

Satz von Thomson / Kelvin

$$\frac{\mathrm{D}\Gamma}{\mathrm{D}t} = -\oint_{S(t)} \left(\frac{1}{\rho}\right) \mathrm{d}p$$

Reibungsfrei,
Barotrop / Inkompressibel,
U existiert

Biot-Savart'sches Gesetz (positiv: ↺, RHR)

Vektorform

$$\mathrm{d}\underline{v} = \frac{\Gamma}{4\pi} \cdot \frac{\underline{a} \times \mathrm{d}\underline{s}}{a^3}$$

Betragsform

$$\mathrm{d}v = \frac{\Gamma}{4\pi} \cdot \frac{\sin(\alpha)\,\mathrm{d}s}{a^2}$$

Einseiting Unendlich

$$v = \frac{\Gamma}{4\pi r} \cdot (1 + \cos(\alpha_1))$$

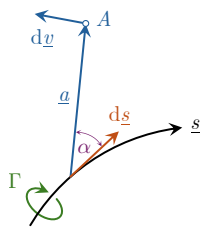
$$\alpha_2 = \pi$$

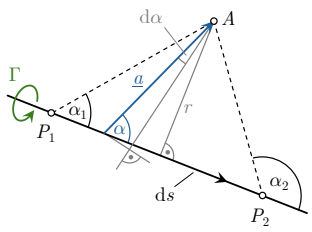
Gerade Wirbellinie

$$\mathrm{d}v = \frac{\Gamma}{4\pi r} \cdot \sin(\alpha)\,\mathrm{d}\alpha$$

Beidseitig Unendlich

$$v = \frac{\Gamma}{2\pi r}$$





arctan2 - Funktion

$$\mathrm{arctan2}(x, y) = \begin{cases} \tan^{-1}\left(\frac{y}{x}\right) & \text{für } x > 0 \\ \tan^{-1}\left(\frac{y}{x}\right) + \pi & \text{für } x < 0, y > 0 \\ \pm\pi & \text{für } x < 0, y = 0 \\ \tan^{-1}\left(\frac{y}{x}\right) - \pi & \text{für } x < 0, y < 0 \\ +\frac{\pi}{2} & \text{für } x = 0, y > 0 \\ -\frac{\pi}{2} & \text{für } x = 0, y < 0 \\ \text{undefiniert} & \text{für } x = 0, y = 0 \end{cases}$$

Rechenregeln

$$\mathrm{arctan2}(x, -y) = -\mathrm{arctan2}(x, y)$$

$$\mathrm{arctan2}(-x, y) = \begin{cases} -\mathrm{arctan2}(-x, y) + \mathrm{sign}(y) \cdot \pi & \text{für } y \neq 0 \\ 0 & \text{für } y = 0 \wedge x < 0 \\ \pm\pi & \text{für } y = 0 \wedge x > 0 \end{cases}$$

(Die Funktion $\mathrm{sign}(x)$ gibt das Vorzeichen von x zurück.)

Potentialtheorie

Geschwindigkeitspotential

Definition

$$\underline{v} = \nabla(\phi)$$

Inkompressibel

$$\Delta(\phi) = 0$$

Grundgleichung

$$\frac{\partial \phi}{\partial t} + \frac{1}{2}\left(\nabla(\phi)\right)^2 + \frac{r}{\rho} + U = F(t)$$

Zirkulation

$$\Gamma_{1\to 2} = \int_1^2 \underline{v} \cdot \mathrm{d}\underline{s} = \phi_2 - \phi_1$$

Laplace-Gleichung (ausgeschrieben)

Kartesisch

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0$$

Zylinderkoordinaten

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \phi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 \phi}{\partial \varphi^2} + \frac{\partial^2 \phi}{\partial z^2} = 0$$

Randbedingungen

Feste Wand

$$\frac{\partial \phi}{\partial n} = 0$$

Bewegte Wand

$$\frac{\partial \phi}{\partial n} = \underline{u} \cdot \underline{n}$$

Im Unendlichen

$$\nabla(\phi) = \underline{v}_\infty$$

Ebene Geschwindigkeiten

Kartesisch

$$u = \frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y}$$

$$v = \frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x}$$

Zylindrisch

$$u_r = \frac{\partial \phi}{\partial r} = \frac{1}{r} \cdot \frac{\partial \psi}{\partial \varphi}$$

$$v_\varphi = \frac{1}{r} \cdot \frac{\partial \phi}{\partial \varphi} = -\frac{\partial \psi}{\partial r}$$

Geschwindigkeitsbeziehungen

Eben (Cauchy-Riemann)

$$u = \frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y}$$

$$v = \frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x}$$

Rotationssymmetrisch

$$v_r = \frac{\partial \phi}{\partial r} = -\frac{1}{r} \cdot \frac{\partial \psi}{\partial z}$$

$$v_z = \frac{\partial \phi}{\partial z} = \frac{1}{r} \cdot \frac{\partial \psi}{\partial r}$$

Resultierende Geschw.

$$F(z) = F(x+iy) = \phi(x,y) + i\psi(x,y)$$

Geschwindigkeiten

$$w = u + iv$$

$$w_* = u - iv = \frac{\mathrm{d}F(z)}{\mathrm{d}z}$$

Druckverteilung

$$p = p_0 - \frac{\rho}{2} a^2 r^{2(n-1)}$$

Ebene Translationsströmung

$$F(z) = a \cdot z = (a_1 - i a_2) \cdot z$$

$$\phi = a_1 \cdot x + a_2 \cdot y$$

$$\psi = a_1 \cdot y - a_2 \cdot x$$

$$u = a_1 \; ; \quad v = a_2$$

Ebene Eckenströmung

$$F(z) = \frac{a}{n} z^n = \frac{a}{n} r^n (\cos(n\varphi) + i \sin(n\varphi))$$

$$\phi = \tfrac{a}{n} r^n \cos(n\varphi)$$

$$\psi = \tfrac{a}{n} r^n \sin(n\varphi)$$

$$|w| = \left|a \cdot z^{n-1}\right| = |a| \cdot r^{n-1}$$

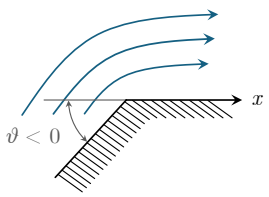
Feste Wände

$$\varphi_{k,n} = k \cdot \frac{\pi}{n}$$

$$\varepsilon = \frac{\pi}{n}$$

$$\vartheta = \pi - \varepsilon = \pi \cdot \frac{n-1}{n}$$

$1/2 < n < 1$

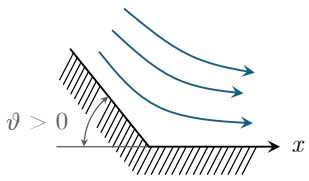


Konvexe Ecke

$$|w| \sim \frac{1}{r^{n-1}}$$

$r = 0 \Rightarrow$ Eckenstr. mit $|w| \to \infty$

$1 < n < 2$

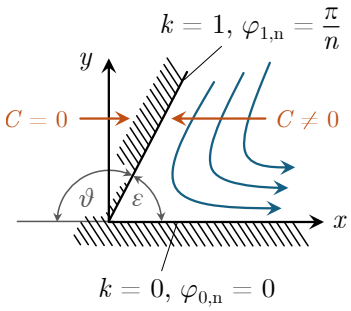


Konkave Ecke

$$|w| \sim r^{n-1}$$

$r = 0 \Rightarrow$ Staupunkt

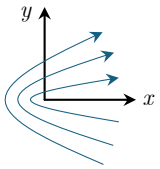
$n > 2$



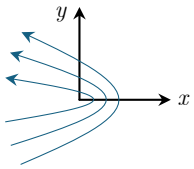
$n = 1/2$

$$F(z) = 2 \cdot a \cdot \sqrt{z} = 2 \cdot a \cdot \sqrt{r} \cdot \left(\cos\left(\frac{\varphi}{2}\right) + i \sin\left(\frac{\varphi}{2}\right)\right)$$

$a =$ reell:



$a =$ imaginär:

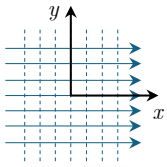


Ebene Randumströmung

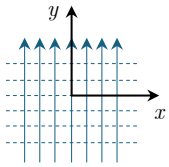
$n = 1$

$$F(z) = a \cdot z$$

$a =$ reell:



$a =$ imaginär:

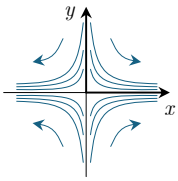


Ebene Translationsströmung

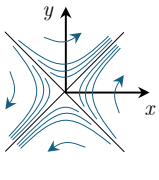
$n = 2$

$$F(z) = \frac{a}{2} \cdot z^2$$

$a =$ reell:



$a =$ imaginär:



Ebene Staupunktströmung

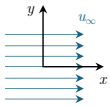
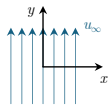
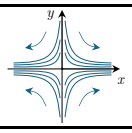
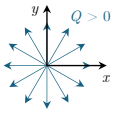
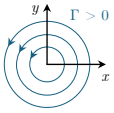
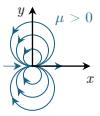
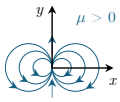
Strömungsbild bestimmen

$\Psi(x,y)$ → $u(x,y), v(x,y)$ → $\left.\begin{matrix} u = 0 \\ v = 0 \end{matrix}\right\} \Rightarrow$ Staupunkte $x_{0,i},\ y_{0,i}$ → $\Psi_{0,i} = \Psi\left(x_{0,i},\ y_{0,i}\right)$ → $\Psi_{\mathrm{S},i}\left(x,y\right) = \Psi_{0,i}$

Gleichungsnummern aus der Vorlesung von Prof. Dr.-Ing. Krämer in Blau.
CAS-Programmaufrufe und ggf. Nummer zur Formelwahl in Orange.

Potentialtheorie

Singularitätentabelle

Bezeichnung	Stromlinienbild	Komplexe Potentialfunktion	Skalare: Potentialfunktion	Skalare: Stromfunktion	Geschwindigkeitskomponenten			
		$F(z)$	$\Phi(x,y)$	$\Psi(x,y)$	$u(x,y)$	$v(x,y)$	$v_r(x,y)$	$v_\varphi(x,y)$
		$F(r,\varphi)$	$\Phi(r,\varphi)$	$\Psi(r,\varphi)$	$u(r,\varphi)$	$v(r,\varphi)$	$v_r(r,\varphi)$	$v_\varphi(r,\varphi)$
Translations-Strömung in x-Richtung		$u_\infty\, z$	$u_\infty\, x$	$u_\infty\, y$	u_∞	0	$u_\infty \frac{x}{\sqrt{x^2+y^2}}$	$-u_\infty \frac{y}{\sqrt{x^2+y^2}}$
		$u_\infty\, r\, e^{i\varphi}$	$u_\infty\, r\, \cos(\varphi)$	$u_\infty\, r\, \sin(\varphi)$	u_∞	0	$u_\infty\, \cos(\varphi)$	$-u_\infty\, \sin(\varphi)$
Translations-Strömung in y-Richtung		$-i\, v_\infty\, z$	$v_\infty\, y$	$-v_\infty\, x$	0	v_∞	$v_\infty \frac{y}{\sqrt{x^2+y^2}}$	$v_\infty \frac{x}{\sqrt{x^2+y^2}}$
		$-i\, v_\infty\, r\, e^{i\varphi}$	$v_\infty\, r\, \sin(\varphi)$	$-v_\infty\, r\, \cos(\varphi)$	0	v_∞	$v_\infty\, \sin(\varphi)$	$v_\infty\, \cos(\varphi)$
Staupunkt- & Eckenstr. $a \in \mathbb{R}^+$		$\frac{a}{2}\, z^2$	$\frac{a}{2}\left(x^2-y^2\right)$	$a\, x\, y$	$a\, x$	$-a\, y$	$a \frac{x^2-y^2}{\sqrt{x^2+y^2}}$	$-a \frac{2\, x\, y}{\sqrt{x^2+y^2}}$
		$\frac{a}{2}\, r^2\, e^{2i\varphi}$	$\frac{a}{2}\, r^2\, \cos(2\,\varphi)$	$\frac{a}{2}\, r^2\, \sin(2\,\varphi)$	$a\, r\, \cos(\varphi)$	$-a\, r\, \sin(\varphi)$	$a\, r\, \cos(2\,\varphi)$	$-a\, r\, \sin(2\,\varphi)$
Quelle, Senke (Ergiebigkeit sei $Q \neq 0$)		$\frac{Q}{2\pi}\, \ln(z)$	$\frac{Q}{2\pi}\, \ln\left(\sqrt{x^2+y^2}\right)$	$\frac{Q}{2\pi}\, \arctan\left(\frac{y}{x}\right)$	$\frac{Q}{2\pi}\, \frac{x}{x^2+y^2}$	$\frac{Q}{2\pi}\, \frac{y}{x^2+y^2}$	$\frac{Q}{2\pi}\, \frac{1}{\sqrt{x^2+y^2}}$	0
		$\frac{Q}{2\pi}\, (\ln(r) + i\,\varphi)$	$\frac{Q}{2\pi}\, \ln(r)$	$\frac{Q}{2\pi}\, \varphi$	$\frac{Q}{2\pi}\, \frac{\cos(\varphi)}{r}$	$\frac{Q}{2\pi}\, \frac{\sin(\varphi)}{r}$	$\frac{Q}{2\pi}\, \frac{1}{r}$	0
Potentialwirbel (Zirkulation sei $\Gamma \neq 0$)		$-i\, \frac{\Gamma}{2\pi}\, \ln(z)$	$\frac{\Gamma}{2\pi}\, \arctan\left(\frac{y}{x}\right)$	$-\frac{\Gamma}{2\pi}\, \ln\left(\sqrt{x^2+y^2}\right)$	$-\frac{\Gamma}{2\pi}\, \frac{y}{x^2+y^2}$	$\frac{\Gamma}{2\pi}\, \frac{x}{x^2+y^2}$	0	$\frac{\Gamma}{2\pi}\, \frac{1}{\sqrt{x^2+y^2}}$
		$-\frac{\Gamma}{2\pi}\, (i\, \ln(r) - \varphi)$	$\frac{\Gamma}{2\pi}\, \varphi$	$-\frac{\Gamma}{2\pi}\, \ln(r)$	$-\frac{\Gamma}{2\pi}\, \frac{\sin(\varphi)}{r}$	$\frac{\Gamma}{2\pi}\, \frac{\cos(\varphi)}{r}$	0	$\frac{\Gamma}{2\pi}\, \frac{1}{r}$
Dipol (Auf der x-Achse)		$-\frac{\mu}{2\pi}\, \frac{1}{z}$	$-\frac{\mu}{2\pi}\, \frac{x}{x^2+y^2}$	$\frac{\mu}{2\pi}\, \frac{y}{x^2+y^2}$	$\frac{\mu}{2\pi}\, \frac{x^2-y^2}{(x^2+y^2)^2}$	$\frac{\mu}{2\pi}\, \frac{2\, x\, y}{(x^2+y^2)^2}$	$\frac{\mu}{2\pi}\, \frac{x}{\sqrt{(x^2+y^2)^3}}$	$\frac{\mu}{2\pi}\, \frac{y}{\sqrt{(x^2+y^2)^3}}$
		$-\frac{\mu}{2\pi}\, \frac{1}{r}\, e^{-i\,\varphi}$	$-\frac{\mu}{2\pi}\, \frac{\cos(\varphi)}{r}$	$\frac{\mu}{2\pi}\, \frac{\sin(\varphi)}{r}$	$\frac{\mu}{2\pi}\, \frac{\cos(2\varphi)}{r^2}$	$\frac{\mu}{2\pi}\, \frac{\sin(2\varphi)}{r^2}$	$\frac{\mu}{2\pi}\, \frac{\cos(\varphi)}{r^2}$	$\frac{\mu}{2\pi}\, \frac{\sin(\varphi)}{r^2}$
Dipol (Auf der y-Achse)		$-i\, \frac{\mu}{2\pi}\, \frac{1}{z}$	$-\frac{\mu}{2\pi}\, \frac{y}{x^2+y^2}$	$-\frac{\mu}{2\pi}\, \frac{x}{x^2+y^2}$	$\frac{\mu}{2\pi}\, \frac{2\, x\, y}{(x^2+y^2)^2}$	$-\frac{\mu}{2\pi}\, \frac{x^2-y^2}{(x^2+y^2)^2}$	$\frac{\mu}{2\pi}\, \frac{y}{\sqrt{(x^2+y^2)^3}}$	$-\frac{\mu}{2\pi}\, \frac{x}{\sqrt{(x^2+y^2)^3}}$
		$-i\, \frac{\mu}{2\pi}\, \frac{1}{r}\, e^{-i\,\varphi}$	$-\frac{\mu}{2\pi}\, \frac{\sin(\varphi)}{r}$	$-\frac{\mu}{2\pi}\, \frac{\cos(\varphi)}{r}$	$\frac{\mu}{2\pi}\, \frac{\sin(2\varphi)}{r^2}$	$-\frac{\mu}{2\pi}\, \frac{\cos(2\varphi)}{r^2}$	$\frac{\mu}{2\pi}\, \frac{\sin(\varphi)}{r^2}$	$-\frac{\mu}{2\pi}\, \frac{\cos(\varphi)}{r^2}$

Dipolmoment $M = -\mu \quad \left[\frac{\mathrm{m}^3}{\mathrm{s}}\right]$ Ergiebigkeit einer Quelle / Senke $E = Q \quad \left[\frac{\mathrm{m}^2}{\mathrm{s}}\right]$

Kreiszylinder

Kutta-Joukowskischer Auftriebssatz

$$F_y = F_\mathrm{A} = b \cdot \rho \cdot u_\infty \cdot \Gamma$$

$\Rightarrow$ Nur Wirbel trägt zum Auftrieb bei

Druckverteilung

$$\underbrace{c_\mathrm{p} = \frac{p_\mathrm{w}(\varphi) - p_\infty}{\frac{\rho}{2} \cdot u_\infty^2} = 1 - \left(\frac{v_{\varphi\mathrm{w}}}{u_\infty}\right)^2}_{c_{\mathrm{p,\,max}} = 1} = \underbrace{1 - \left(4\, \sin^2(\varphi) + \frac{2\,\Gamma\, \sin(\varphi)}{\pi\, a\, u_\infty} + \left(\frac{\Gamma}{2\,\pi\, a\, u_\infty}\right)^2\right)}_{c_{\mathrm{p,\,min}} = -3}$$

Inkompressibel

Einfache Staupunkte

Staupunkte eines Dipol-Zylinders mit Radius a:

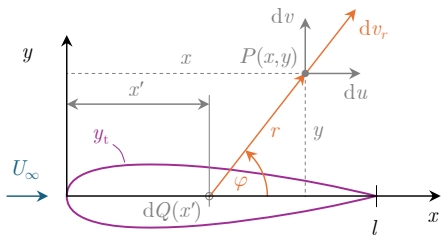
$$\sin(\varphi_\mathrm{s}) = -\frac{\Gamma}{4\,\pi\, u_\infty\, a}$$

Druckbeiwert Allg.

$$c_\mathrm{p} = \frac{p - p_\infty}{\frac{\rho}{2} \cdot u_\infty^2}$$

Tropfentheorie

Geschwindigkeiten



Allgemein

$$U(x,y) = U_\infty + \frac{1}{2\pi} \int_0^l q(x') \, \frac{x - x'}{(x - x')^2 + y^2} \, \mathrm{d}x'$$

$$V(x,y) = \frac{1}{2\pi} \int_0^l q(x') \, \frac{y}{(x - x')^2 + y^2} \, \mathrm{d}x'$$

Dünnes Profil (Körperoberfläche)

$$U(x) = U_\infty \left(1 + \frac{1}{\pi} \oint_0^l \frac{\frac{\mathrm{d}y_\mathrm{t}}{\mathrm{d}x}(x')}{x - x'} \, \mathrm{d}x' \right)$$

$$V(x) = \frac{q(x)}{2}$$

Kinematische Verträglichkeit

$$\frac{\mathrm{d}y_\mathrm{t}}{\mathrm{d}x} = \frac{V(x)}{U(x)}$$

Druckverteilung

$$c_\mathrm{p} = 1 - \frac{U^2 + V^2}{U_\infty^2}$$

Quell- und Senkenverteilung

Allgemein

$$q(x) = 2 \, \frac{\mathrm{d}}{\mathrm{d}x} \left((U_\infty + u) \cdot y_\mathrm{t} \right)$$

Dünnes Profil

$$q(x) = 2 \, U_\infty \, \frac{\mathrm{d}y_\mathrm{t}(x)}{\mathrm{d}x}$$

Riegels-Korrektur

Druckverteilung

$$c_\mathrm{p} = 1 - \left(\frac{U_\mathrm{k}}{U_\infty} \right)^2$$

Definition

$$\kappa(x) = \frac{1}{\cos(\theta)} = \sqrt{1 + \tan^2(\theta)} = \sqrt{1 + \left(\frac{\mathrm{d}y_\mathrm{t}}{\mathrm{d}x} \right)^2}$$

Geschwindigkeit

$$U_\mathrm{k}(x) = \frac{U}{\kappa(x)} = \frac{U_\infty}{\kappa(x)} \left(1 + \frac{1}{\pi} \oint_0^l \frac{\frac{\mathrm{d}y_\mathrm{t}}{\mathrm{d}x}(x')}{x - x'} \, \mathrm{d}x' \right)$$

Fourier-Ansatz

Definition

$$Y_\mathrm{t} = \frac{1}{2} \sum_{\nu=1}^{n} b_\nu \sin(\nu \, \varphi)$$

$$X = \frac{1}{2} \left(1 + \cos(\varphi) \right)$$

Geschwindigkeit

$$U_\mathrm{k}(\varphi) = \frac{U_\infty}{\kappa(\varphi)} \left(1 + \sum_{\nu=1}^{n} \nu \, b_\nu \, \frac{\sin(\nu \, \varphi)}{\sin(\varphi)} \right)$$

Riegels-Faktor

$$\kappa(\varphi) = \sqrt{1 + \left(\frac{\mathrm{d}Y_\mathrm{t}}{\mathrm{d}X} \right)^2} = \sqrt{1 + \left(\frac{\mathrm{d}Y_\mathrm{t}}{\mathrm{d}X} \, \frac{\mathrm{d}\varphi}{\mathrm{d}X} \right)^2}$$

Gleichungsnummern aus der Vorlesung von Prof. Dr.-Ing. Krämer in Blau.
CAS-Programmaufrufe und ggf. Nummer zur Formelwahl in Orange.

Skelettheorie

Geschwindigkeiten

Allgemein

$$u(x,y) = \frac{1}{2\,\pi} \int_0^1 \gamma(x') \, \frac{y - y_\mathrm{s}(x')}{(x-x')^2 + (y - y_\mathrm{s}(x'))^2} \, \mathrm{d}x'$$

$$v(x,y) = -\frac{1}{2\,\pi} \int_0^1 \gamma(x') \, \frac{x - x'}{(x-x')^2 + (y - y_\mathrm{s}(x'))^2} \, \mathrm{d}x'$$

Wirbel auf der x-Achse

Gesamt

$$U(X) = U_\infty + \frac{\gamma(X)}{2} \, \mathrm{sign}(Y - Y_\mathrm{s})$$

$$V(X) = U_\infty \, \alpha - \frac{1}{2\,\pi} \oint_0^l \frac{\gamma(X')}{X - X'} \, \mathrm{d}X'$$

Induziert in y-Richtung

$$v(X) = -\frac{1}{2\,\pi} \oint_0^l \frac{\gamma(X')}{X - X'} \, \mathrm{d}X'$$

Auftrieb

$$A = \rho \, U_\infty \, b \, \Gamma$$

Moment

$$M = -b \, \rho \, U_\infty \int_0^l \gamma(x) \, x \, \mathrm{d}x$$

Druck

Oberseite

$$p_\mathrm{o}(x) = p_\infty - \frac{\rho}{2} \, U_\infty \, \gamma(x)$$

Unterseite

$$p_\mathrm{u}(x) = p_\infty + \frac{\rho}{2} \, U_\infty \, \gamma(x)$$

Druckbeiwert

$$\Delta C_\mathrm{p} = 2 \, \frac{\gamma(x)}{U_\infty}$$

Auftriebsbeiwert

$$C_\mathrm{a} = \frac{2}{U_\infty} \int_0^1 \gamma(X) \, \mathrm{d}X = \int_0^1 \Delta C_\mathrm{p}(X) \, \mathrm{d}X$$

Momentenbeiwert

$$C_\mathrm{m} = -\frac{2}{U_\infty} \int_0^1 \gamma(X) \, X \, \mathrm{d}X = -\int_0^1 \Delta C_\mathrm{p}(X) \, X \, \mathrm{d}X$$

Skelettlinie

$$Y_\mathrm{s}(X) = \alpha \, X - \frac{1}{2\,\pi} \int_0^1 \frac{\gamma(X')}{U_\infty} \, \ln \left| \frac{X - X'}{X'} \right| \mathrm{d}X'$$

$$\alpha = \frac{1}{2\,\pi} \int_0^1 \frac{\gamma(X')}{U_\infty} \, \ln \left| \frac{1 - X'}{X'} \right| \mathrm{d}X'$$

Ansatz von Birnbaum-Ackermann und Glauert

Zirkulationsverteilung

$$\gamma(\varphi) = 2 \, U_\infty \left(A_0 \, \tan\left(\frac{\varphi}{2}\right) + \sum_{n=1}^{\infty} A_n \, \sin(n \, \varphi) \right)$$

$$A_0 = \alpha - \frac{1}{\pi} \int_0^\pi \frac{\mathrm{d}Y_\mathrm{s}(X)}{\mathrm{d}X} \, \mathrm{d}\varphi$$

$$A_n = -\frac{2}{\pi} \int_0^\pi \frac{\mathrm{d}Y_\mathrm{s}(X)}{\mathrm{d}X} \, \cos(n \, \varphi) \, \mathrm{d}\varphi$$

Induzierte Geschwindigkeit

$$\frac{v(\varphi)}{U_\infty} = -\left(A_0 + \sum_{n=1}^{N} A_n \, \cos(n \, \varphi) \right)$$

Kinematische Strömungsbedingung

$$\alpha - \frac{\mathrm{d}Y_\mathrm{s}(X)}{\mathrm{d}X} = A_0 + \sum_{n=1}^{N} A_n \, \cos(n \, \varphi)$$

Auftriebsbeiwert

$$C_\mathrm{a} = \pi \, (2 \, A_0 + A_1)$$

Momentenbeiwert

$$C_\mathrm{m} = -\frac{\pi}{4} \, (2 \, A_0 + 2 \, A_1 + A_2)$$

Druckpunktlage

$$\frac{x_\mathrm{A}}{l} = -\frac{C_\mathrm{m}}{C_\mathrm{a}}$$

Überlagerung

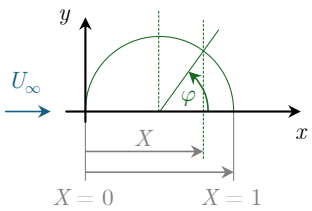
$$\frac{U}{U_\infty} = 1 + \left(\frac{U_1}{U_\infty} - 1\right) + \left(\frac{U_2}{U_\infty} - 1\right) + \left(\frac{U_3}{U_\infty} - 1\right)$$

Bedingung: $\frac{\alpha}{\pi}, \frac{f}{l}, \frac{d}{l} \ll 1$

Skelettlinie

$$Y_\mathrm{s} = \alpha \, X - A_0 \, X - \int_0^X A_0 \, \cos(n \, \varphi) \, \mathrm{d}X$$

Transformation



$$X = \frac{1}{2} \, (1 + \cos(\varphi))$$

Energiesatz

Thermische Zustandsgleichung

$$p\,V = m\,R\,T \qquad p = \rho\,R\,T$$

Enthalpie

$$h = e + p\,v = c_\mathrm{p}\,T$$

Innere Energie

$$e = c_\mathrm{v}\,T$$

Entropie

$$\mathrm{d}s = \frac{\mathrm{d}q_\mathrm{rev}}{T} + \mathrm{d}s_\mathrm{irrev}$$

1. Hauptsatz

$$\mathrm{d}e = \mathrm{d}q + \mathrm{d}w = T\,\mathrm{d}s - p\,\mathrm{d}v$$

$$\mathrm{d}q = \mathrm{d}h - v\,\mathrm{d}p$$

Maxwell-Beziehung

$$\left.\frac{\partial e}{\partial v}\right|_T = T\,\left.\frac{\partial p}{\partial T}\right|_v - p$$

Spezifische Wärmekapazität

Konst. Volumen

$$c_\mathrm{v} = \left.\frac{\partial e}{\partial T}\right|_v = \frac{R}{\kappa - 1}$$

Konst. Druck

$$c_\mathrm{p} = \left.\frac{\partial h}{\partial T}\right|_p = R\,\frac{\kappa}{\kappa - 1}$$

Zusammenhang

$$R = c_\mathrm{p} - c_\mathrm{v}$$

Isentropenexponent

$$\kappa = \frac{c_\mathrm{p}}{c_\mathrm{v}}$$

Einatomig: $\kappa = \frac{5}{3}$

Zweiatomig: $\kappa = \frac{7}{5}$

Entropieänderung

$$\Delta s = R\,\ln\left(\frac{\rho_1}{\rho_2}\,\left(\frac{T_2}{T_1}\right)^{\frac{1}{\kappa-1}}\right)$$

Isentropenbeziehungen

$$\frac{p}{\rho^\kappa} = \text{konst.} \qquad \frac{\rho_2}{\rho_1} = \left(\frac{T_2}{T_1}\right)^{\frac{1}{\kappa-1}} \qquad \frac{p_2}{p_1} = \left(\frac{\rho_2}{\rho_1}\right)^\kappa = \left(\frac{T_2}{T_1}\right)^{\frac{\kappa}{\kappa-1}}$$

Energiegleichung

Nicht-Konservativ

$$\rho\,\frac{\mathrm{D}}{\mathrm{D}t}\left(e + \frac{V^2}{2}\right) = \rho\,\dot{q} + \frac{\partial}{\partial x}\left(k\,\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k\,\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial z}\left(k\,\frac{\partial T}{\partial z}\right) - \frac{\partial(u\,p)}{\partial x} - \frac{\partial(v\,p)}{\partial y} - \frac{\partial(w\,p)}{\partial z}$$
$$+ \frac{\partial(u\,\tau_{xx})}{\partial x} + \frac{\partial(u\,\tau_{yx})}{\partial y} + \frac{\partial(u\,\tau_{zx})}{\partial z} + \frac{\partial(u\,\tau_{xy})}{\partial x} + \frac{\partial(u\,\tau_{yy})}{\partial y} + \frac{\partial(u\,\tau_{zy})}{\partial z} + \frac{\partial(u\,\tau_{xz})}{\partial x} + \frac{\partial(u\,\tau_{yz})}{\partial y} + \frac{\partial(u\,\tau_{zz})}{\partial z} + \rho\,\underline{f}\cdot\underline{V}$$

Konservativ

$$\frac{\partial}{\partial t}\left(\rho\,\left(e + \frac{V^2}{2}\right)\right) + \nabla\cdot\left(\rho\,\left(e + \frac{V^2}{2}\right)\,\underline{V}\right) = \rho\,\dot{q} + \frac{\partial}{\partial x}\left(k\,\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k\,\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial z}\left(k\,\frac{\partial T}{\partial z}\right) - \frac{\partial(u\,p)}{\partial x} - \frac{\partial(v\,p)}{\partial y} - \frac{\partial(w\,p)}{\partial z}$$
$$+ \frac{\partial(u\,\tau_{xx})}{\partial x} + \frac{\partial(u\,\tau_{yx})}{\partial y} + \frac{\partial(u\,\tau_{zx})}{\partial z} + \frac{\partial(u\,\tau_{xy})}{\partial x} + \frac{\partial(u\,\tau_{yy})}{\partial y} + \frac{\partial(u\,\tau_{zy})}{\partial z} + \frac{\partial(u\,\tau_{xz})}{\partial x} + \frac{\partial(u\,\tau_{yz})}{\partial y} + \frac{\partial(u\,\tau_{zz})}{\partial z} + \rho\,\underline{f}\cdot\underline{V}$$

Energiegleichung der laminaren Grenzschicht

Allgemein:

$$\rho\,u\,\frac{\partial}{\partial x}\left(c_\mathrm{p}\,T + \frac{u^2}{2}\right) + \rho\,v\,\frac{\partial}{\partial y}\left(c_\mathrm{p}\,T + \frac{u^2}{2}\right) = \frac{\partial}{\partial y}\left(k\,\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial y}\left(\eta\,u\,\frac{\partial u}{\partial y}\right)$$

Für den Fall $\mathrm{Pr} = 1$

Definition

$$\rho\,u\,\frac{\partial}{\partial x}\left(c_\mathrm{p}\,T + \frac{u^2}{2}\right) + \rho\,v\,\frac{\partial}{\partial y}\left(c_\mathrm{p}\,T + \frac{u^2}{2}\right) = \frac{\partial}{\partial y}\left(\eta\,\frac{\partial}{\partial y}\left(c_\mathrm{p}\,T + \frac{u^2}{2}\right)\right)$$

$\frac{\mathrm{d}p}{\mathrm{d}x} = \text{beliebig}$

$$T + \frac{u^2}{2\,c_\mathrm{p}} = T_{0\infty}$$

$$\Rightarrow r = 1$$

$\frac{\mathrm{d}p}{\mathrm{d}x} = 0$

$$T(u) = T_\mathrm{W} + \frac{u}{u_\mathrm{e}}\,(T_{0\mathrm{e}} - T_\mathrm{W}) - \underbrace{\frac{u_\mathrm{e}^2}{2\,c_\mathrm{p}}\,\left(\frac{u}{u_\mathrm{e}}\right)^2}_{\approx\,0\text{ für }\mathrm{Ma}\,\ll\,1}$$

Recovery-Faktor

$$r = \frac{2}{(\kappa - 1)\,\mathrm{Ma}_\mathrm{e}^2}\,\left(\frac{T_\mathrm{aW}}{T_\mathrm{e}} - 1\right) \approx \begin{cases} \sqrt{\mathrm{Pr}} & \text{für laminare Grenzschicht} \\ \sqrt[3]{\mathrm{Pr}} & \text{für turbulente Grenzschicht} \end{cases}$$

Recovery-Temperatur

$$T_\mathrm{r} = T_\mathrm{e} + r\,\frac{u_\mathrm{e}^2}{2\,c_\mathrm{p}}$$

Prandtl-Zahl

$$\mathrm{Pr} = \frac{\eta\,c_\mathrm{p}}{k}$$

Reynolds-Analogie

Definition

$$\dot{q}_\mathrm{W} = c_\mathrm{p}\,\frac{T_\mathrm{r} - T_\mathrm{W}}{u_\mathrm{e}}\,\tau_\mathrm{W}$$

(8.89) ↩

Stanton-Zahl

$$\mathrm{St} = \frac{\dot{q}_\mathrm{W}}{\rho_\mathrm{e}\,u_\mathrm{e}\,c_\mathrm{p}\,(T_\mathrm{r} - T_\mathrm{W})} \approx \begin{cases} \frac{c_\mathrm{f}}{2} & \text{für } \mathrm{Pr} = 1 \\ \frac{c_\mathrm{f}}{2}\,\mathrm{Pr}^{-\frac{2}{3}} & \text{für } \mathrm{Pr} < 0.5 \end{cases}$$

(8.91) ↩ (8.92) / (8.93) ↩

$$c_\mathrm{f} = \frac{2\,\tau_\mathrm{W}}{\rho_\mathrm{e}\,u_\mathrm{e}^2} = \frac{0.664}{\sqrt{\mathrm{Re}_x}}$$

(8.90) ↩ (8.94) ↩

$$\mathrm{Re}_x = \frac{\rho\,u_\mathrm{e}\,x}{\eta}$$

Energiegleichung der turbulenten Grenzschicht ($\mathrm{Ma} \ll 1$)

Allgemein

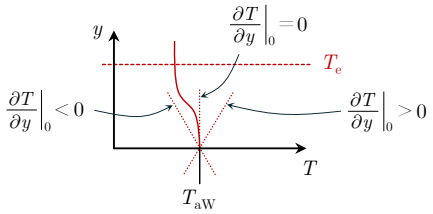
$$\rho\,\overline{u}\,\frac{\partial\overline{T}}{\partial x} - \rho\,\overline{v}\,\frac{\partial\overline{T}}{\partial y} = \frac{k}{c_\mathrm{p}}\,\frac{\partial^2\overline{T}}{\partial y^2} - \rho\,\frac{\partial\left(\overline{v'\,T'}\right)}{\partial y}$$

Lösung

$$\overline{T} = T_\mathrm{W} + (T_{0\mathrm{e}} - T_\mathrm{W})\,\frac{\overline{u}}{u_\mathrm{e}}$$

(8.106) ↩

Temperatur-Gradient



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Gasdynamik

Massenerhaltung

$$\frac{\partial(\rho A)}{\partial t} + \frac{\partial(\rho A V)}{\partial x} = 0 \quad (9.4b)$$

Impulssatz

$$\rho \frac{\mathrm{D}V}{\mathrm{D}t} = -\frac{\partial p}{\partial x} \quad (9.5)$$

Energieerhaltung

$$\frac{\mathrm{D}h_0}{\mathrm{D}t} = \frac{1}{\rho}\frac{\partial p}{\partial t}$$

Entropieänderung

$$\frac{\mathrm{D}s}{\mathrm{D}t} = 0$$

Schallgeschwindigkeit

$$c = \sqrt{\kappa R T} = \sqrt{\kappa \frac{p}{\rho}}$$

Stationäre Stromröhre

Massenerhaltung

$$\dot{m} = \rho A V = \text{konst.}$$

Impulssatz

$$\frac{V^2}{2} + \int \frac{1}{\rho}\,\mathrm{d}p = \text{konst.}$$

Energiegleichung

$$h_0 = h + \frac{V^2}{2} = \text{konst.}$$

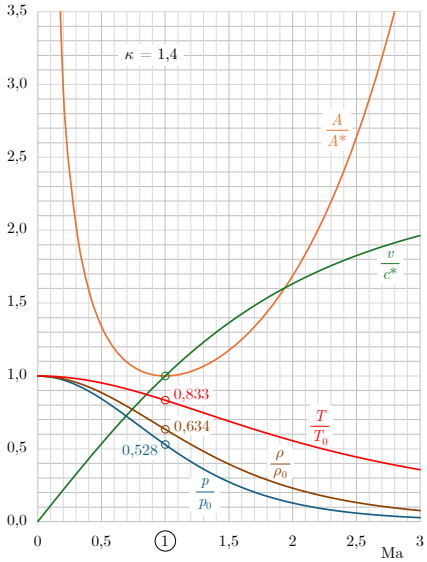
Kritische Schallgeschwindigkeit

$$c_\mathrm{L} = c^* = \sqrt{\frac{2}{\kappa+1}}\,c_0 = \sqrt{2\,\frac{\kappa}{\kappa+1}\,R T_0}$$

(9.22a) ↛

$$\left(\frac{V}{c^*}\right)^2 = \frac{\kappa+1}{\kappa-1}\left(1 - \frac{T}{T_0}\right) = \frac{\kappa+1}{\frac{2}{\mathrm{Ma}^2} + \kappa - 1}$$

(9.21b) ↛



Adiabate Strömungsvorgänge

astrr() astrkr()

Ruhezustand

$$\frac{T_0}{T} = 1 + \frac{\kappa-1}{2}\,\mathrm{Ma}^2 \quad (9.19)$$

$$\frac{p_0}{p} = \left(\frac{T_0}{T}\right)^{\frac{\kappa}{\kappa-1}} = \left(1 + \frac{\kappa-1}{2}\,\mathrm{Ma}^2\right)^{\frac{\kappa}{\kappa-1}}$$

(9.20a) ↛

$$\frac{\rho_0}{\rho} = \left(\frac{T_0}{T}\right)^{\frac{1}{\kappa-1}} = \left(1 + \frac{\kappa-1}{2}\,\mathrm{Ma}^2\right)^{\frac{1}{\kappa-1}}$$

(9.20b) ↛

$$h_0 = h + \frac{V^2}{2} \Leftrightarrow T_0 = T + \frac{V^2}{2\,c_\mathrm{p}} \quad (8.60)$$

Krit. Zustand (Ma = 1)

$$\frac{T^*}{T_0} = \frac{2}{\kappa+1}$$

$$\frac{p^*}{p_0} = \left(\frac{2}{\kappa+1}\right)^{\frac{\kappa}{\kappa-1}}$$

$$\frac{\rho^*}{\rho_0} = \left(\frac{2}{\kappa+1}\right)^{\frac{1}{\kappa-1}}$$

↚ (9.26)

$$\frac{A}{A^*} = \frac{1}{\mathrm{Ma}}\left[\frac{2}{\kappa+1}\left(1 + \frac{\kappa-1}{2}\mathrm{Ma}^2\right)\right]^{\frac{\kappa+1}{2\,(\kappa-1)}}$$

mstr()

$$\dot{m} = \frac{\mathrm{Ma}}{\left(1 + \frac{\kappa-1}{2}\,\mathrm{Ma}^2\right)^{\frac{\kappa+1}{2\,(\kappa-1)}}}\sqrt{\frac{\kappa}{R T_0}}\,p_0\,A = \left(\frac{2}{\kappa+1}\right)^{\frac{\kappa+1}{2\,(\kappa-1)}}\sqrt{\frac{\kappa}{R T_0}}\,p_0\,A^* \quad (9.24)$$

Machzahl

$$\mathrm{Ma} = \frac{V}{c}$$

Lavalzahl

la() (9.23b) ↚

$$\mathrm{La} = \frac{V}{c^*} = \sqrt{\frac{\kappa+1}{2 + (\kappa-1)\,\mathrm{Ma}^2}}\,\mathrm{Ma}$$

(9.23a) ↛

$$\mathrm{La}_\mathrm{max} = \sqrt{\frac{\kappa+1}{\kappa-1}}$$

Luft: $\mathrm{La}_\mathrm{max} = 2,449$

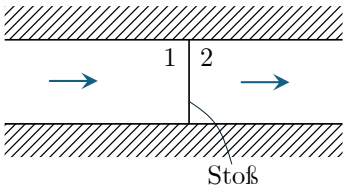
Stationäre senkrechte Verdichtungsstöße

Erhaltungssätze

$$\rho_1 V_1 = \rho_2 V_2$$

$$\rho_1 V_1^2 + p_1 = \rho_2 V_2^2 + p_2$$

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2}$$



Entropiezunahme

$$s_2 - s_1 = R\,\ln\left(\frac{\rho_1}{\rho_2}\left(\frac{T_2}{T_1}\right)^{\frac{1}{\kappa-1}}\right)$$

$$= R\,\ln\left(\frac{p_{01}}{p_{02}}\right)$$

(9.40) ↛

Stoßbeziehungen

ssv()

① $$\frac{p_2}{p_1} = \frac{2\,\kappa\,\mathrm{Ma}_1^2 - \kappa + 1}{\kappa+1} \quad (9.47a)$$

② $$\frac{\rho_2}{\rho_1} = \frac{(\kappa+1)\,\mathrm{Ma}_1^2}{2 + (\kappa-1)\,\mathrm{Ma}_1^2} \quad (9.38b)$$

(9.38a) ↚

③ $$\frac{T_2}{T_1} = \frac{h_2}{h_1} = \frac{V_2^2}{V_1^2} = \frac{(2\,\kappa\,\mathrm{Ma}_1^2 - \kappa + 1)(2 + (\kappa-1)\,\mathrm{Ma}_1^2)}{(\kappa+1)^2\,\mathrm{Ma}_1^2}$$

④ $$\mathrm{Ma}_2 = \sqrt{\frac{(\kappa-1)(\mathrm{Ma}_1^2 - 1) + \kappa + 1}{2\,\kappa\,(\mathrm{Ma}_1^2 - 1) + \kappa + 1}} \quad (9.35)$$

⑤ $$\frac{p_{02}}{p_{01}} = \left[\frac{\left(\frac{\kappa+1}{2}\right)^{\kappa+1}\mathrm{Ma}_1^{(2\,\kappa)}}{\left(1 + \frac{\kappa-1}{2}\,\mathrm{Ma}_1^2\right)^{\kappa}\left(\kappa\,\mathrm{Ma}_1^2 - \frac{\kappa-1}{2}\right)}\right]^{\frac{1}{\kappa-1}} \quad (9.39b)$$

Rayleigh-Pitot-Formel

rpf()

$$\frac{p_\mathrm{pitot}}{p_1} = \left[\frac{\left(\frac{\kappa+1}{2}\right)^{\kappa+1}\mathrm{Ma}_1^{(2\,\kappa)}}{\kappa\,\mathrm{Ma}_1^2 - \frac{\kappa-1}{2}}\right]^{\frac{1}{\kappa-1}} \quad (9.41)$$

Rankine-Hugoniot-Beziehung

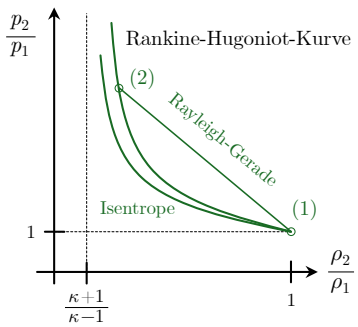
$$e_2 - e_1 = \frac{p_1 + p_2}{2}\,(v_1 - v_2)$$

↚ (9.53) (9.54) ↚

$$\frac{p_2}{p_1} = \frac{(\kappa+1)\,\rho_2 - (\kappa-1)\,\rho_1}{(\kappa+1)\,\rho_1 - (\kappa-1)\,\rho_2}$$

Rayleigh-Gerade

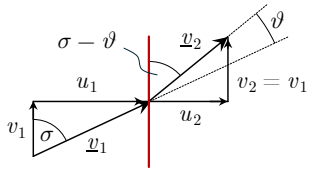
$$\frac{p_2 - p_1}{v_2 - v_1} = -\left(\frac{u_1}{v_1}\right)^2$$



Gasdynamik

Schräge Verdichtungsstöße

Stoßbeziehungen sv()



(9.43) ↷

$$\mathrm{Ma}_{1n} = \mathrm{Ma}_1 \sin(\sigma)$$

$$\mathrm{Ma}_{2n} = \mathrm{Ma}_2 \sin(\sigma - \vartheta)$$

(9.47a) ↷

$$\frac{p_2}{p_1} = 1 + \frac{2\,\kappa}{\kappa+1}\left(\mathrm{Ma}_1^2 \sin^2(\sigma) - 1\right)$$

$$\frac{\rho_2}{\rho_1} = \frac{u_1}{u_2} = \frac{(\kappa+1)\,\mathrm{Ma}_1^2 \sin^2(\sigma)}{2 + (\kappa-1)\,\mathrm{Ma}_1^2 \sin^2(\sigma)}$$

(9.44) ↗

$$\frac{T_2}{T_1} = \frac{V_2^2}{V_1^2} = \frac{p_2}{p_1}\,\frac{\rho_1}{\rho_2} = \frac{(2\,\kappa\,\mathrm{Ma}_1^2 \sin^2(\sigma) - \kappa + 1)\,(2 + (\kappa-1)\,\mathrm{Ma}_1^2 \sin^2(\sigma))}{(\kappa+1)^2\,\mathrm{Ma}_1^2 \sin^2(\sigma)}$$

$$\frac{\mathrm{Ma}_1}{\mathrm{Ma}_2} = \left[\frac{1 + \frac{2\,\kappa}{\kappa+1}\left(\mathrm{Ma}_1^2 \sin^2(\sigma) - 1\right)}{1 - \frac{2}{\kappa+1}\left(1 - \frac{1}{\mathrm{Ma}_1^2 \sin^2(\sigma)}\right)}\right]^{\frac{1}{2}} \frac{\sin(\sigma - \vartheta)}{\sin(\sigma)} \qquad (9.59)$$

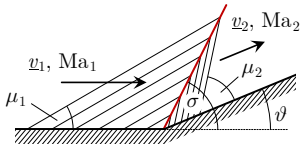
$$\frac{p_{02}}{p_{01}} = \left[\frac{\left(\frac{\kappa+1}{2}\right)^{\kappa+1} (\mathrm{Ma}_1 \sin(\sigma))^{2\kappa}}{\left(1 + \frac{\kappa-1}{2}(\mathrm{Ma}_1 \sin(\sigma))^2\right)^{\kappa} \left(\kappa\,(\mathrm{Ma}_1 \sin(\sigma))^2 - \frac{\kappa-1}{2}\right)}\right]^{\frac{1}{\kappa-1}} \qquad (9.58)$$

Mach'sche Welle

(9.67) ↷

$$\mu = \arcsin\left(\frac{1}{\mathrm{Ma}}\right)$$

Es gilt: $\mu \le \sigma \le \frac{\pi}{2}$



Umlenkwinkel

$$\tan(\vartheta) = \cot(\sigma)\,\frac{\mathrm{Ma}_1^2 \sin^2(\sigma) - 1}{1 + \left(\frac{\kappa+1}{2} - \sin^2(\sigma)\right)\mathrm{Ma}_1^2} \qquad (9.49)$$

Entropiezunahme (Ideales Gas)

$$s_2 - s_1 = c_\mathrm{v} \ln\left(\frac{p_2}{p_1}\left(\frac{\rho_1}{\rho_2}\right)^{\kappa}\right) \qquad (9.47c)$$

Schräge Stoßexpansion schs()

$$\frac{T_2}{T_1} = \frac{1 + \frac{\kappa-1}{2}\mathrm{Ma}_1^2}{1 + \frac{\kappa-1}{2}\mathrm{Ma}_2^2}$$

(9.77a) ↗

$$\frac{p2}{p_1} = \left(\frac{1 + \frac{\kappa-1}{2}\mathrm{Ma}_1^2}{1 + \frac{\kappa-1}{2}\mathrm{Ma}_2^2}\right)^{\frac{\kappa}{\kappa-1}}$$

(9.77b) ↗

Öffnungskegel

$$\delta = \mu_2 - \mu_1 + \Delta\vartheta_{12}$$

$$\nu(\mathrm{Ma}_2) = \nu(\mathrm{Ma}_1) + \vartheta_{12}$$

(9.75) ↗

Prandtl-Meyer-Funktion pmf()

$$\nu(\mathrm{Ma}) = \sqrt{\frac{\kappa+1}{\kappa-1}}\,\arctan\left(\sqrt{\frac{\kappa-1}{\kappa+1}\left(\mathrm{Ma}^2 - 1\right)}\right) - \arctan\left(\sqrt{\mathrm{Ma}^2 - 1}\right)$$

(9.74) ↗ $\quad (\nu_{\max} = 130{,}454°)$

Instationäre eindimensionale Strömung

Kolbengeschwindigkeit

$$\frac{V_\mathrm{k}'}{c_1} = \frac{2}{\kappa+1}\,\frac{\mathrm{Ma}_\mathrm{s}^2 - 1}{\mathrm{Ma}_\mathrm{s}}$$

Machzahl

$$\mathrm{Ma}_2' = \frac{\mathrm{Ma}_\mathrm{s}^2 - 1}{\sqrt{\left(1 + \frac{\kappa-1}{2}\mathrm{Ma}_\mathrm{s}^2\right)\left(\kappa\,\mathrm{Ma}_\mathrm{s}^2 - \frac{\kappa-1}{2}\right)}}$$

Stoßbeziehungen

$$\frac{p_2'}{p_1'} = \frac{2\,\kappa\,\mathrm{Ma}_\mathrm{s}^2 - \kappa + 1}{\kappa+1}$$

$$\frac{p_{02}'}{p_1'} = \left(\frac{\kappa+1}{2\,\kappa\,\mathrm{Ma}_\mathrm{s}^2 - \kappa + 1}\right)^{\frac{1}{\kappa-1}} \left(\mathrm{Ma}_\mathrm{s}^2\,\frac{2\,(\kappa-1)\,\mathrm{Ma}_\mathrm{s}^2 + (3-\kappa)}{2 + (\kappa-1)\,\mathrm{Ma}_\mathrm{s}^2}\right)^{\frac{\kappa}{\kappa-1}}$$

$$\frac{T_2'}{T_1'} = \frac{(2\,\kappa\,\mathrm{Ma}_\mathrm{s}^2 - \kappa + 1)\,(2 + (\kappa-1)\,\mathrm{Ma}_\mathrm{s}^2)}{(\kappa+1)^2\,\mathrm{Ma}_\mathrm{s}^2}$$

$$\frac{T_{02}'}{T_1} = \frac{2}{\kappa+1}\left((\kappa-1)\,\mathrm{Ma}_\mathrm{s}^2 + \frac{3-\kappa}{2}\right)$$

$\mathrm{Ma}_\mathrm{s} > 7$

$$\mathrm{Ma}_2 \approx \sqrt{\frac{\kappa-1}{2\,\kappa}}$$

$$\frac{p_{02}}{p_{01}} \approx \left(\frac{\kappa+1}{2\,\kappa}\left(\frac{\kappa+1}{\kappa-1}\,\kappa\right)\frac{1}{\mathrm{Ma}_\mathrm{s}^2}\right)^{\frac{1}{\kappa-1}}$$

$$\frac{T_2}{T_1} \approx 2\,\kappa\,\frac{(\kappa-1)}{(\kappa+1)^2}\,\mathrm{Ma}_\mathrm{s}^2$$

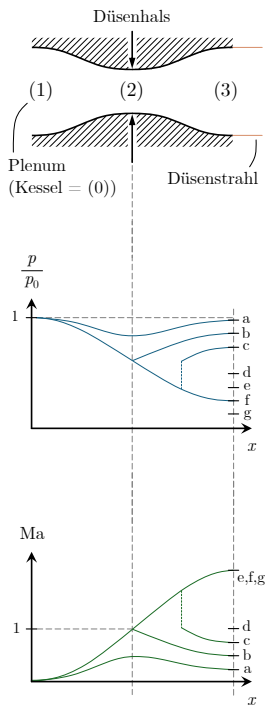
$$\mathrm{Ma}_2' \approx \sqrt{\frac{2}{\kappa\,(\kappa-1)}}$$

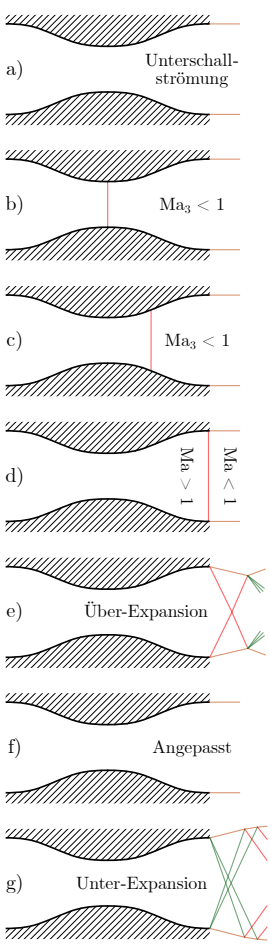
$$\frac{p_{02}'}{p_{01}'} = \frac{p_{02}'}{p_1'} = \frac{p_{02}'}{p_1} \approx \left(\frac{\kappa+1}{\kappa}\right)^{\frac{1}{\kappa-1}} \mathrm{Ma}_\mathrm{s}^2$$

$$\frac{T_{02}'}{T_{01}'} = \frac{T_{02}'}{T_1'} = \frac{T_{02}'}{T_1} \approx 2\,\frac{(\kappa-1)}{(\kappa+1)}\,\mathrm{Ma}_\mathrm{s}^2$$

$$\frac{p_2}{p_1} = \frac{p_2'}{p_1'} \approx \frac{2\,\kappa}{\kappa+1}\,\mathrm{Ma}_\mathrm{s}^2$$

Düsenströmung





Gleichungsnummern aus der Vorlesung von Prof. Dr.-Ing. Krämer in Blau.
CAS-Programmaufrufe und ggf. Nummer zur Formelwahl in Orange.

Anhang A1

Umrechnungstabelle isentrope Strömung, Unterschall (Ideales Gas; $\kappa = 1,4$)

Ma	La	$\frac{p}{p_0}$	$\frac{\rho}{\rho_0}$	$\frac{T}{T_0}$	$\frac{A}{A^*}$
0.00	0.0000	1.00000	1.00000	1.00000	∞
0.01	0.0110	0.99993	0.99995	0.99998	57.87384
0.02	0.0219	0.99972	0.99980	0.99992	28.94213
0.03	0.0329	0.99937	0.99955	0.99982	19.30054
0.04	0.0438	0.99888	0.99920	0.99968	14.48149
0.05	0.0548	0.99825	0.99875	0.99950	11.59144
0.06	0.0657	0.99748	0.99820	0.99928	9.66591
0.07	0.0766	0.99658	0.99755	0.99902	8.29153
0.08	0.0876	0.99553	0.99681	0.99872	7.26161
0.09	0.0985	0.99435	0.99596	0.99838	6.46134
0.10	0.1094	0.99303	0.99502	0.99800	5.82183
0.11	0.1204	0.99158	0.99398	0.99759	5.29923
0.12	0.1313	0.98998	0.99284	0.99713	4.86342
0.13	0.1422	0.98826	0.99160	0.99663	4.49686
0.14	0.1531	0.98640	0.99027	0.99610	4.18240
0.15	0.1639	0.98441	0.98884	0.99552	3.91034
0.16	0.1748	0.98228	0.98731	0.99491	3.67274
0.17	0.1857	0.98003	0.98569	0.99425	3.46351
0.18	0.1965	0.97765	0.98398	0.99356	3.27793
0.19	0.2074	0.97514	0.98218	0.99283	3.11226
0.20	0.2182	0.97250	0.98028	0.99206	2.96352
0.21	0.2290	0.96973	0.97829	0.99126	2.82929
0.22	0.2398	0.96685	0.97620	0.99041	2.70760
0.23	0.2506	0.96383	0.97403	0.98953	2.59681
0.24	0.2614	0.96070	0.97177	0.98861	2.49556
0.25	0.2722	0.95745	0.96942	0.98765	2.40271
0.26	0.2829	0.95408	0.96698	0.98666	2.31729
0.27	0.2936	0.95060	0.96446	0.98563	2.23847
0.28	0.3043	0.94700	0.96185	0.98456	2.16555
0.29	0.3150	0.94329	0.95916	0.98346	2.09793
0.30	0.3257	0.93947	0.95638	0.98232	2.03507
0.31	0.3364	0.93554	0.95352	0.98114	1.97651
0.32	0.3470	0.93150	0.95058	0.97993	1.92185
0.33	0.3576	0.92736	0.94756	0.97868	1.87074
0.34	0.3682	0.92312	0.94446	0.97740	1.82288
0.35	0.3788	0.91877	0.94128	0.97609	1.77797
0.36	0.3893	0.91433	0.93803	0.97473	1.73578
0.37	0.3999	0.90979	0.93470	0.97335	1.69609
0.38	0.4104	0.90516	0.93130	0.97193	1.65870
0.39	0.4209	0.90043	0.92782	0.97048	1.62343
0.40	0.4313	0.89561	0.92427	0.96899	1.59014
0.41	0.4418	0.89071	0.92066	0.96747	1.55867
0.42	0.4522	0.88572	0.91697	0.96592	1.52891
0.43	0.4626	0.88065	0.91322	0.96434	1.50072
0.44	0.4729	0.87550	0.90940	0.96272	1.47401
0.45	0.4833	0.87027	0.90551	0.96108	1.44867
0.46	0.4936	0.86496	0.90157	0.95940	1.42463
0.47	0.5038	0.85958	0.89756	0.95769	1.40180
0.48	0.5141	0.85413	0.89349	0.95595	1.38010
0.49	0.5243	0.84861	0.88936	0.95418	1.35947
0.50	0.5345	0.84302	0.88517	0.95238	1.33984
0.51	0.5447	0.83737	0.88093	0.95055	1.32117
0.52	0.5548	0.83165	0.87663	0.94869	1.30339
0.53	0.5649	0.82588	0.87228	0.94681	1.28645
0.54	0.5750	0.82005	0.86788	0.94489	1.27032
0.55	0.5851	0.81417	0.86342	0.94295	1.25495
0.56	0.5951	0.80823	0.85892	0.94098	1.24029
0.57	0.6051	0.80224	0.85437	0.93898	1.22633
0.58	0.6150	0.79621	0.84978	0.93696	1.21301
0.59	0.6249	0.79013	0.84514	0.93491	1.20031
0.60	0.6348	0.78400	0.84045	0.93284	1.18820
0.61	0.6447	0.77784	0.83573	0.93073	1.17665
0.62	0.6545	0.77164	0.83096	0.92861	1.16565
0.63	0.6643	0.76540	0.82616	0.92646	1.15515
0.64	0.6740	0.75913	0.82132	0.92428	1.14515
0.65	0.6837	0.75283	0.81644	0.92208	1.13562
0.66	0.6934	0.74650	0.81153	0.91986	1.12654
0.67	0.7031	0.74014	0.80659	0.91762	1.11789
0.68	0.7127	0.73376	0.80162	0.91535	1.10965
0.69	0.7223	0.72735	0.79661	0.91306	1.10182
0.70	0.7318	0.72093	0.79158	0.91075	1.09437
0.71	0.7413	0.71448	0.78652	0.90841	1.08729
0.72	0.7508	0.70803	0.78143	0.90606	1.08057
0.73	0.7602	0.70155	0.77632	0.90369	1.07419
0.74	0.7696	0.69507	0.77119	0.90129	1.06814
0.75	0.7789	0.68857	0.76604	0.89888	1.06242
0.76	0.7883	0.68207	0.76086	0.89644	1.05700
0.77	0.7975	0.67556	0.75567	0.89399	1.05188
0.78	0.8068	0.66905	0.75046	0.89152	1.04705
0.79	0.8160	0.66254	0.74523	0.88903	1.04251
0.80	0.8251	0.65602	0.73999	0.88652	1.03823
0.81	0.8343	0.64951	0.73474	0.88400	1.03422
0.82	0.8433	0.64300	0.72947	0.88146	1.03046
0.83	0.8524	0.63650	0.72419	0.87890	1.02696
0.84	0.8614	0.63000	0.71891	0.87633	1.02370
0.85	0.8704	0.62351	0.71361	0.87374	1.02067
0.86	0.8793	0.61703	0.70831	0.87114	1.01787
0.87	0.8882	0.61057	0.70300	0.86852	1.01530
0.88	0.8970	0.60412	0.69768	0.86589	1.01294
0.89	0.9058	0.59768	0.69236	0.86324	1.01080
0.90	0.9146	0.59126	0.68704	0.86059	1.00886
0.91	0.9233	0.58486	0.68172	0.85791	1.00713
0.92	0.9320	0.57848	0.67640	0.85523	1.00560
0.93	0.9406	0.57211	0.67108	0.85253	1.00426
0.94	0.9493	0.56578	0.66576	0.84982	1.00311
0.95	0.9578	0.55946	0.66044	0.84710	1.00215
0.96	0.9663	0.55317	0.65513	0.84437	1.00136
0.97	0.9748	0.54691	0.64982	0.84162	1.00076
0.98	0.9832	0.54067	0.64452	0.83887	1.00034
0.99	0.9916	0.53446	0.63923	0.83611	1.00008

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Anhang A2.1

Umrechnungstabelle isentrope Strömung, Überschall (Ideales Gas; $\kappa = 1,4$)

Ma	La	$\frac{p}{p_0}$	$\frac{\rho}{\rho_0}$	$\frac{T}{T_0}$	$\frac{A}{A^*}$	μ	ν
1.00	1.0000	.52828	.63394	.83333	1.00000	90.000	0.000
1.01	1.0083	.52213	.62866	.83055	1.00008	81.931	0.045
1.02	1.0166	.51602	.62339	.82776	1.00033	78.635	0.126
1.03	1.0248	.50994	.61813	.82496	1.00074	76.138	0.229
1.04	1.0330	.50389	.61289	.82215	1.00131	74.058	0.351
1.05	1.0411	.49787	.60765	.81934	1.00203	72.247	0.487
1.06	1.0492	.49189	.60243	.81651	1.00291	70.630	0.637
1.07	1.0573	.48595	.59723	.81368	1.00394	69.160	0.797
1.08	1.0653	.48005	.59203	.81085	1.00512	67.808	0.968
1.09	1.0733	.47418	.58686	.80800	1.00645	66.553	1.148
1.10	1.0812	.46835	.58170	.80515	1.00793	65.380	1.336
1.11	1.0891	.46257	.57655	.80230	1.00955	64.277	1.532
1.12	1.0970	.45682	.57143	.79944	1.01131	63.235	1.735
1.13	1.1048	.45112	.56632	.79657	1.01322	62.246	1.944
1.14	1.1126	.44545	.56123	.79370	1.01527	61.306	2.160
1.15	1.1203	.43983	.55616	.79083	1.01745	60.408	2.381
1.16	1.1280	.43425	.55112	.78795	1.01978	59.550	2.607
1.17	1.1356	.42872	.54609	.78506	1.02224	58.727	2.839
1.18	1.1432	.42322	.54108	.78218	1.02484	57.936	3.074
1.19	1.1508	.41778	.53610	.77929	1.02757	57.176	3.314
1.20	1.1583	.41238	.53114	.77640	1.03044	56.443	3.558
1.21	1.1658	.40702	.52620	.77350	1.03344	55.735	3.806
1.22	1.1732	.40171	.52129	.77061	1.03657	55.052	4.057
1.23	1.1806	.39645	.51640	.76771	1.03983	54.391	4.312
1.24	1.1879	.39123	.51154	.76481	1.04323	53.751	4.569
1.25	1.1952	.38606	.50670	.76190	1.04675	53.130	4.830
1.26	1.2025	.38093	.50189	.75900	1.05041	52.528	5.093
1.27	1.2097	.37586	.49710	.75610	1.05419	51.943	5.359
1.28	1.2169	.37083	.49234	.75319	1.05810	51.375	5.627
1.29	1.2240	.36585	.48761	.75029	1.06214	50.823	5.898
1.30	1.2311	.36091	.48290	.74738	1.06630	50.285	6.170
1.31	1.2382	.35603	.47822	.74448	1.07060	49.761	6.445
1.32	1.2452	.35119	.47358	.74158	1.07502	49.251	6.721
1.33	1.2522	.34640	.46895	.73867	1.07957	48.753	7.000
1.34	1.2591	.34166	.46436	.73577	1.08424	48.268	7.279
1.35	1.2660	.33697	.45980	.73287	1.08904	47.795	7.561
1.36	1.2729	.33233	.45526	.72997	1.09396	47.332	7.843
1.37	1.2797	.32773	.45076	.72707	1.09902	46.880	8.128
1.38	1.2864	.32319	.44628	.72418	1.10419	46.439	8.413
1.39	1.2932	.31869	.44184	.72128	1.10950	46.007	8.699
1.40	1.2999	.31424	.43742	.71839	1.11493	45.585	8.987
1.41	1.3065	.30984	.43304	.71550	1.12048	45.171	9.276
1.42	1.3131	.30549	.42869	.71262	1.12616	44.767	9.565
1.43	1.3197	.30119	.42436	.70973	1.13197	44.371	9.855
1.44	1.3262	.29693	.42007	.70685	1.13790	43.983	10.146
1.45	1.3327	.29272	.41581	.70398	1.14396	43.603	10.438
1.46	1.3392	.28856	.41158	.70111	1.15015	43.230	10.730
1.47	1.3456	.28445	.40739	.69824	1.15646	42.865	11.023
1.48	1.3520	.28039	.40322	.69537	1.16290	42.507	11.317
1.49	1.3583	.27637	.39909	.69251	1.16947	42.155	11.611

Ma	La	$\frac{p}{p_0}$	$\frac{\rho}{\rho_0}$	$\frac{T}{T_0}$	$\frac{A}{A^*}$	μ	ν
1.50	1.3646	.27240	.39498	.68966	1.17617	41.810	11.905
1.51	1.3708	.26848	.39091	.68680	1.18299	41.472	12.200
1.52	1.3770	.26461	.38688	.68396	1.18994	41.140	12.495
1.53	1.3832	.26078	.38287	.68112	1.19702	40.813	12.790
1.54	1.3894	.25700	.37890	.67828	1.20423	40.493	13.086
1.55	1.3955	.25326	.37496	.67545	1.21157	40.178	13.381
1.56	1.4015	.24957	.37105	.67262	1.21904	39.868	13.677
1.57	1.4075	.24593	.36717	.66980	1.22664	39.564	13.973
1.58	1.4135	.24233	.36332	.66699	1.23438	39.265	14.269
1.59	1.4195	.23878	.35951	.66418	1.24224	38.971	14.564
1.60	1.4254	.23527	.35573	.66138	1.25023	38.682	14.860
1.61	1.4313	.23181	.35198	.65858	1.25836	38.398	15.156
1.62	1.4371	.22839	.34827	.65579	1.26663	38.118	15.452
1.63	1.4429	.22501	.34458	.65301	1.27502	37.843	15.747
1.64	1.4487	.22168	.34093	.65023	1.28355	37.572	16.043
1.65	1.4544	.21840	.33731	.64746	1.29222	37.305	16.338
1.66	1.4601	.21515	.33372	.64470	1.30102	37.043	16.633
1.67	1.4657	.21195	.33017	.64194	1.30996	36.784	16.928
1.68	1.4713	.20879	.32664	.63919	1.31904	36.530	17.222
1.69	1.4769	.20567	.32315	.63645	1.32825	36.279	17.516
1.70	1.4825	.20259	.31969	.63371	1.33761	36.032	17.810
1.71	1.4880	.19956	.31626	.63099	1.34710	35.789	18.103
1.72	1.4935	.19656	.31287	.62827	1.35673	35.549	18.396
1.73	1.4989	.19361	.30950	.62556	1.36651	35.312	18.689
1.74	1.5043	.19070	.30617	.62285	1.37643	35.080	18.981
1.75	1.5097	.18782	.30287	.62016	1.38649	34.850	19.273
1.76	1.5150	.18499	.29959	.61747	1.39670	34.624	19.565
1.77	1.5203	.18219	.29635	.61479	1.40705	34.400	19.855
1.78	1.5256	.17944	.29315	.61212	1.41755	34.180	20.146
1.79	1.5308	.17672	.28997	.60945	1.42819	33.963	20.436
1.80	1.5360	.17404	.28682	.60680	1.43898	33.749	20.725
1.81	1.5411	.17140	.28370	.60415	1.44992	33.538	21.014
1.82	1.5463	.16879	.28061	.60151	1.46101	33.329	21.302
1.83	1.5514	.16622	.27756	.59888	1.47225	33.124	21.590
1.84	1.5564	.16369	.27453	.59626	1.48365	32.921	21.877
1.85	1.5614	.16120	.27153	.59365	1.49519	32.720	22.163
1.86	1.5664	.15873	.26857	.59104	1.50689	32.523	22.449
1.87	1.5714	.15631	.26563	.58845	1.51875	32.328	22.734
1.88	1.5763	.15392	.26272	.58586	1.53076	32.135	23.019
1.89	1.5812	.15156	.25984	.58329	1.54293	31.945	23.303
1.90	1.5861	.14924	.25699	.58072	1.55526	31.757	23.586
1.91	1.5909	.14695	.25417	.57816	1.56774	31.571	23.869
1.92	1.5957	.14470	.25138	.57561	1.58039	31.388	24.151
1.93	1.6005	.14247	.24861	.57307	1.59320	31.207	24.432
1.94	1.6052	.14028	.24588	.57054	1.60617	31.028	24.712
1.95	1.6099	.13813	.24317	.56802	1.61931	30.852	24.992
1.96	1.6146	.13600	.24049	.56551	1.63261	30.677	25.271
1.97	1.6192	.13390	.23784	.56301	1.64608	30.505	25.549
1.98	1.6239	.13184	.23521	.56051	1.65972	30.335	25.827
1.99	1.6284	.12981	.23262	.55803	1.67352	30.166	26.104

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Anhang A2.2

Umrechnungstabelle isentrope Strömung, Überschall (Ideales Gas; $\kappa = 1,4$)

Ma	La	$\frac{p}{p_0}$	$\frac{\rho}{\rho_0}$	$\frac{T}{T_0}$	$\frac{A}{A^*}$	μ	ν
2.00	1.6330	.12780	.23005	.55556	1.68750	30.000	26.380
2.01	1.6375	.12583	.22751	.55309	1.70165	29.836	26.655
2.02	1.6420	.12389	.22499	.55064	1.71597	29.673	26.930
2.03	1.6465	.12197	.22250	.54819	1.73047	29.512	27.203
2.04	1.6509	.12009	.22004	.54576	1.74514	29.353	27.476
2.05	1.6553	.11823	.21760	.54333	1.75999	29.196	27.748
2.06	1.6597	.11640	.21519	.54092	1.77502	29.041	28.020
2.07	1.6640	.11460	.21281	.53851	1.79022	28.888	28.290
2.08	1.6683	.11282	.21045	.53611	1.80561	28.736	28.560
2.09	1.6726	.11107	.20811	.53373	1.82119	28.585	28.829
2.10	1.6769	.10935	.20580	.53135	1.83694	28.437	29.097
2.11	1.6811	.10766	.20352	.52898	1.85288	28.290	29.364
2.12	1.6853	.10599	.20126	.52663	1.86901	28.145	29.631
2.13	1.6895	.10434	.19902	.52428	1.88533	28.001	29.896
2.14	1.6936	.10273	.19681	.52194	1.90184	27.859	30.161
2.15	1.6977	.10113	.19463	.51962	1.91854	27.718	30.425
2.16	1.7018	$.99562 \cdot 10^{-1}$	.19247	.51730	1.93543	27.578	30.688
2.17	1.7059	$.98017 \cdot 10^{-1}$	.19033	.51499	1.95252	27.441	30.951
2.18	1.7099	$.96495 \cdot 10^{-1}$	.18821	.51269	1.96981	27.304	31.212
2.19	1.7139	$.94997 \cdot 10^{-1}$	.18612	.51041	1.98729	27.169	31.473
2.20	1.7179	$.93522 \cdot 10^{-1}$	.18405	.50813	2.00497	27.036	31.732
2.21	1.7219	$.92070 \cdot 10^{-1}$	.18200	.50586	2.02286	26.903	31.991
2.22	1.7258	$.90640 \cdot 10^{-1}$	.17998	.50361	2.04094	26.773	32.249
2.23	1.7297	$.89232 \cdot 10^{-1}$	.17798	.50136	2.05923	26.643	32.507
2.24	1.7336	$.87846 \cdot 10^{-1}$	.17600	.49912	2.07773	26.515	32.763
2.25	1.7374	$.86482 \cdot 10^{-1}$	.17404	.49689	2.09643	26.388	33.018
2.26	1.7412	$.85139 \cdot 10^{-1}$	.17211	.49468	2.11535	26.262	33.273
2.27	1.7450	$.83817 \cdot 10^{-1}$	.17020	.49247	2.13447	26.138	33.527
2.28	1.7488	$.82515 \cdot 10^{-1}$	.16830	.49027	2.15381	26.014	33.780
2.29	1.7526	$.81234 \cdot 10^{-1}$	.16643	.48809	2.17336	25.892	34.032
2.30	1.7563	$.79973 \cdot 10^{-1}$	.16458	.48591	2.19313	25.771	34.283
2.31	1.7600	$.78731 \cdot 10^{-1}$	.16275	.48374	2.21311	25.652	34.533
2.32	1.7637	$.77509 \cdot 10^{-1}$	.16095	.48158	2.23332	25.533	34.782
2.33	1.7673	$.76306 \cdot 10^{-1}$	.15916	.47944	2.25375	25.416	35.031
2.34	1.7709	$.75122 \cdot 10^{-1}$	.15739	.47730	2.27440	25.300	35.279
2.35	1.7745	$,73957 \cdot 10^{-1}$	.15564	.47517	2.29527	25.184	35.525
2.36	1.7781	$.72810 \cdot 10^{-1}$	.15391	.47306	2.31638	25.070	35.771
2.37	1.7817	$.71681 \cdot 10^{-1}$	.15221	.47095	2.33771	24.957	36.016
2.38	1.7852	$.70570 \cdot 10^{-1}$	.15052	.46885	2.35927	24.845	36.261
2.39	1.7887	$.69476 \cdot 10^{-1}$	.14885	.46676	2.38107	24.734	36.504
2.40	1.7922	$.68400 \cdot 10^{-1}$	.14720	.46468	2.40310	24.624	36.746
2.41	1.7956	$.67340 \cdot 10^{-1}$	.14556	.46262	2.42536	24.515	36.988
2.42	1.7991	$.66297 \cdot 10^{-1}$	.14395	.46056	2.44787	24.407	37.229
2.43	1.8025	$.65271 \cdot 10^{-1}$	.14235	.45851	2.47061	24.301	37.469
2.44	1.8059	$.64261 \cdot 10^{-1}$	.14078	.45647	2.49360	24.195	37.708
2.45	1.8092	$.63267 \cdot 10^{-1}$	.13922	.45444	2.51683	24.090	37.946
2.46	1.8126	$.62289 \cdot 10^{-1}$	.13768	.45242	2.54031	23.985	38.183
2.47	1.8159	$.61326 \cdot 10^{-1}$	.13615	.45041	2.56403	23.882	38.419
2.48	1.8192	$.60378 \cdot 10^{-1}$	.13465	.44841	2.58801	23.780	38.655
2.49	1.8225	$.59446 \cdot 10^{-1}$	.13316	.44642	2.61223	23.679	38.890
2.50	1.8257	$.58528 \cdot 10^{-1}$	.13169	.44444	2.63672	23.578	39.124
2.51	1.8290	$.57625 \cdot 10^{-1}$	.13023	.44247	2.66145	23.479	39.356
2.52	1.8322	$.56736 \cdot 10^{-1}$	.12879	.44051	2.68645	23.380	39.589
2.53	1.8354	$.55861 \cdot 10^{-1}$	.12737	.43856	2.71171	23.282	39.820
2.54	1.8386	$.55000 \cdot 10^{-1}$	.12597	.43662	2.73722	23.185	40.050
2.55	1.8417	$.54153 \cdot 10^{-1}$	.12458	.43469	2.76301	23.089	40.280
2.56	1.8448	$.53319 \cdot 10^{-1}$	.12321	.43277	2.78906	22.993	40.508
2.57	1.8479	$.52499 \cdot 10^{-1}$	.12185	.43085	2.81538	22.899	40.736
2.58	1.8510	$.51692 \cdot 10^{-1}$	.12051	.42895	2.84197	22.805	40.963
2.59	1.8541	$.50897 \cdot 10^{-1}$	.11918	.42706	2.86883	22.712	41.189
2.60	1.8571	$.50115 \cdot 10^{-1}$	.11787	.42517	2.89597	22.620	41.415
2.61	1.8602	$.49346 \cdot 10^{-1}$	.11658	.42330	2.92339	22.528	41.639
2.62	1.8632	$.48589 \cdot 10^{-1}$	.11530	.42143	2.95108	22.438	41.863
2.63	1.8662	$.47844 \cdot 10^{-1}$	.11403	.41957	2.97906	22.348	42.085
2.64	1.8691	$.47110 \cdot 10^{-1}$	.11278	.41773	3.00733	22.259	42.307
2.65	1.8721	$.46389 \cdot 10^{-1}$	.11154	.41589	3.03588	22.170	42.528
2.66	1.8750	$.45679 \cdot 10^{-1}$	.11032	.41406	3.06472	22.082	42.749
2.67	1.8779	$.44980 \cdot 10^{-1}$	.10911	.41224	3.09385	21.995	42.968
2.68	1.8808	$.44293 \cdot 10^{-1}$	.10792	.41043	3.12327	21.909	43.187
2.69	1.8837	$.43616 \cdot 10^{-1}$	.10674	.40863	3.15299	21.823	43.405
2.70	1.8865	$.42950 \cdot 10^{-1}$	.10557	.40684	3.18301	21.738	43.621
2.71	1.8894	$.42295 \cdot 10^{-1}$	.10442	.40505	3.21332	21.654	43.838
2.72	1.8922	$.41650 \cdot 10^{-1}$	.10328	.40328	3.24395	21.571	44.053
2.73	1.8950	$.41016 \cdot 10^{-1}$	.10215	.40151	3.27487	21.488	44.267
2.74	1.8978	$.40391 \cdot 10^{-1}$	.10104	.39976	3.30611	21.405	44.481
2.75	1.9005	$.39777 \cdot 10^{-1}$	$.99939 \cdot 10^{-1}$	.39801	3.33765	21.324	44.694
2.76	1.9033	$.39172 \cdot 10^{-1}$	$.98852 \cdot 10^{-1}$	.39627	3.36951	21.243	44.906
2.77	1.9060	$.38577 \cdot 10^{-1}$	$.97777 \cdot 10^{-1}$	.39454	3.40168	21.162	45.117
2.78	1.9087	$.37992 \cdot 10^{-1}$	$.96714 \cdot 10^{-1}$	.39282	3.43417	21.083	45.327
2.79	1.9114	$.37415 \cdot 10^{-1}$	$.95664 \cdot 10^{-1}$	.39111	3.46698	21.003	45.537
2.80	1.9140	$.36848 \cdot 10^{-1}$	$.94627 \cdot 10^{-1}$	.38941	3.50012	20.925	45.746
2.81	1.9167	$.36290 \cdot 10^{-1}$	$.93601 \cdot 10^{-1}$	.38771	3.53358	20.847	45.954
2.82	1.9193	$.35741 \cdot 10^{-1}$	$.92587 \cdot 10^{-1}$	.38603	3.56736	20.770	46.161
2.83	1.9219	$.35201 \cdot 10^{-1}$	$.91585 \cdot 10^{-1}$	.38435	3.60148	20.693	46.367
2.84	1.9245	$.34669 \cdot 10^{-1}$	$.90595 \cdot 10^{-1}$	.38268	3.63593	20.617	46.573
2.85	1.9271	$.34146 \cdot 10^{-1}$	$.89616 \cdot 10^{-1}$	.38103	3.67071	20.541	46.778
2.86	1.9297	$.33631 \cdot 10^{-1}$	$.88648 \cdot 10^{-1}$	.37937	3.70584	20.466	46.982
2.87	1.9323	$.33124 \cdot 10^{-1}$	$.87692 \cdot 10^{-1}$	.37773	3.74130	20.391	47.185
2.88	1.9348	$.32625 \cdot 10^{-1}$	$.86747 \cdot 10^{-1}$	.37610	3.77711	20.318	47.388
2.89	1.9373	$.32135 \cdot 10^{-1}$	$.85813 \cdot 10^{-1}$	.37447	3.81326	20.244	47.589
2.90	1.9398	$.31652 \cdot 10^{-1}$	$.84889 \cdot 10^{-1}$	.37286	3.84976	20.171	47.790
2.91	1.9423	$.31176 \cdot 10^{-1}$	$.83977 \cdot 10^{-1}$	.37125	3.88662	20.099	47.990
2.92	1.9448	$.30708 \cdot 10^{-1}$	$.83075 \cdot 10^{-1}$	.36965	3.92382	20.027	48.190
2.93	1.9472	$.30248 \cdot 10^{-1}$	$.82184 \cdot 10^{-1}$	.36806	3.96139	19.956	48.388
2.94	1.9497	$.29795 \cdot 10^{-1}$	$.81302 \cdot 10^{-1}$	.36647	3.99931	19.885	48.586
2.95	1.9521	$.29349 \cdot 10^{-1}$	$.80432 \cdot 10^{-1}$	.36490	4.03760	19.815	48.783
2.96	1.9545	$.28910 \cdot 10^{-1}$	$.79571 \cdot 10^{-1}$	.36333	4.07625	19.745	48.980
2.97	1.9569	$.28479 \cdot 10^{-1}$	$.78720 \cdot 10^{-1}$	.36177	4.11527	19.676	49.175
2.98	1.9593	$.28054 \cdot 10^{-1}$	$.77879 \cdot 10^{-1}$	.36022	4.15466	19.607	49.370
2.99	1.9616	$.27635 \cdot 10^{-1}$	$.77048 \cdot 10^{-1}$	.35868	4.19442	19.539	49.564
3.00	1.9640	$.27224 \cdot 10^{-1}$	$.76226 \cdot 10^{-1}$	.35714	4.23457	19.471	49.757
3.10	1.9866	$.23449 \cdot 10^{-1}$	$.68517 \cdot 10^{-1}$	.34223	4.65731	18.819	51.650
3.20	2.0079	$.20228 \cdot 10^{-1}$	$.61654 \cdot 10^{-1}$	.32808	5.12096	18.210	53.470
3.30	2.0278	$.17477 \cdot 10^{-1}$	$.55541 \cdot 10^{-1}$	.31466	5.62865	17.640	55.222
3.40	2.0466	$.15125 \cdot 10^{-1}$	$.50093 \cdot 10^{-1}$	.30193	6.18370	17.105	56.908
3.50	2.0642	$.13111 \cdot 10^{-1}$	$.45233 \cdot 10^{-1}$	.28986	6.78962	16.602	58.530
3.60	2.0808	$.11385 \cdot 10^{-1}$	$.40894 \cdot 10^{-1}$	.27840	7.45011	16.128	60.091
3.70	2.0964	$.99029 \cdot 10^{-2}$	$.37017 \cdot 10^{-1}$	.26752	8.16906	15.680	61.595
3.80	2.1111	$.86290 \cdot 10^{-2}$	$,33549 \cdot 10^{-1}$	.25720	8.95058	15.258	63.044
3.90	2.1250	$.75321 \cdot 10^{-2}$	$.30445 \cdot 10^{-1}$	.24740	9.79897	14.857	64.440
4.00	2.1381	$.65861 \cdot 10^{-2}$	$.27662 \cdot 10^{-1}$	.23810	10.71874	14.478	65.785
4.10	2.1505	$.57690 \cdot 10^{-2}$	$.25164 \cdot 10^{-1}$	.22925	11.71464	14.117	67.082
4.20	2.1622	$.50621 \cdot 10^{-2}$	$.22921 \cdot 10^{-1}$	.22085	12.79163	13.774	68.333
4.30	2.1732	$.44494 \cdot 10^{-2}$	$.20903 \cdot 10^{-1}$	.21286	13.95489	13.448	69.541
4.40	2.1837	$.39176 \cdot 10^{-2}$	$.19087 \cdot 10^{-1}$	.20525	15.20985	13.137	70.706
4.50	2.1936	$.34553 \cdot 10^{-2}$	$.17449 \cdot 10^{-1}$	.19802	16.56218	12.840	71.832
4.60	2.2030	$.30526 \cdot 10^{-2}$	$.15971 \cdot 10^{-1}$	.19113	18.01777	12.556	72.919
4.70	2.2119	$.27012 \cdot 10^{-2}$	$.14635 \cdot 10^{-1}$	.18457	19.58280	12.284	73.970
4.80	2.2204	$.23943 \cdot 10^{-2}$	$.13427 \cdot 10^{-1}$	.17832	21.26368	12.025	74.986
4.90	2.2284	$.21256 \cdot 10^{-2}$	$.12333 \cdot 10^{-1}$	.17235	23.06709	11.776	75.969
5.00	2.2361	$.18900 \cdot 10^{-2}$	$.11340 \cdot 10^{-1}$	.16667	25.00000	11.537	76.920
6.00	2.2953	$.63336 \cdot 10^{-3}$	$.51936 \cdot 10^{-2}$	.12195	53.17979	9.594	84.955
7.00	2.3333	$.24156 \cdot 10^{-3}$	$.26088 \cdot 10^{-2}$	$.92593 \cdot 10^{-1}$	104.14290	8.213	90.973
8.00	2.3591	$.10243 \cdot 10^{-3}$	$.14135 \cdot 10^{-2}$	$.72464 \cdot 10^{-1}$	190.10940	7.181	95.625
9.00	2.3772	$.47386 \cdot 10^{-4}$	$.81504 \cdot 10^{-3}$	$.58140 \cdot 10^{-1}$	327.18940	6.379	99.318
10.00	2.3905	$.23563 \cdot 10^{-4}$	$.49482 \cdot 10^{-3}$	$.47619 \cdot 10^{-1}$	535.93770	5.739	102.316
∞	2.4495	0	0	0	1	0	130.454

Anhang A3

Umrechnungstabelle senkrechter Stoß, Überschall (Ideales Gas; $\kappa = 1,4$)

Ma_1	Ma_2	$\frac{p_2}{p_1}$	$\frac{\rho_2}{\rho_1}$	$\frac{T_2}{T_1}$	$\frac{p_{02}}{p_{01}}$
2.00	.57735	4.50000	2.66667	1.68750	.72087
2.01	.57565	4.54678	2.68145	1.69564	.71620
2.02	.57397	4.59380	2.69618	1.70382	.71153
2.03	.57232	4.64105	2.71085	1.71203	.70685
2.04	.57068	4.68853	2.72546	1.72027	.70218
2.05	.56906	4.73625	2.74001	1.72855	.69751
2.06	.56747	4.78419	2.75451	1.73686	.69284
2.07	.56589	4.83238	2.76895	1.74520	.68817
2.08	.56433	4.88080	2.78332	1.75359	.68351
2.09	.56280	4.92944	2.79764	1.76200	.67886
2.10	.56128	4.97833	2.81190	1.77045	.67420
2.11	.55978	5.02744	2.82610	1.77893	.66956
2.12	.55829	5.07679	2.84024	1.78745	.66492
2.13	.55683	5.12638	2.85432	1.79601	.66029
2.14	.55538	5.17619	2.86834	1.80459	.65567
2.15	.55395	5.22624	2.88231	1.81322	.65105
2.16	.55254	5.27653	2.89621	1.82187	.64645
2.17	.55115	5.32704	2.91005	1.83057	.64185
2.18	.54977	5.37779	2.92383	1.83930	.63727
2.19	.54840	5.42878	2.93756	1.84806	.63270
2.20	.54706	5.47999	2.95122	1.85686	.62814
2.21	.54572	5.53144	2.96482	1.86569	.62359
2.22	.54441	5.58313	2.97836	1.87456	.61905
2.23	.54311	5.63504	2.99185	1.88347	.61453
2.24	.54182	5.68719	3.00527	1.89241	.61002
2.25	.54055	5.73958	3.01863	1.90138	.60553
2.26	.53930	5.79219	3.03193	1.91040	.60105
2.27	.53805	5.84504	3.04518	1.91944	.59659
2.28	.53683	5.89813	3.05836	1.92853	.59214
2.29	.53561	5.95144	3.07148	1.93764	.58771
2.30	.53441	6.00499	3.08455	1.94680	.58330
2.31	.53322	6.05878	3.09755	1.95599	.57890
2.32	.53205	6.11279	3.11049	1.96522	.57452
2.33	.53089	6.16704	3.12338	1.97448	.57015
2.34	.52974	6.22153	3.13620	1.98378	.56581
2.35	.52861	6.27624	3.14896	1.99311	.56148
2.36	.52749	6.33119	3.16167	2.00248	.55718
2.37	.52638	6.38638	3.17431	2.01189	.55289
2.38	.52528	6.44179	3.18690	2.02133	.54862
2.39	.52419	6.49744	3.19943	2.03081	.54437
2.40	.52312	6.55333	3.21189	2.04033	.54014
2.41	.52206	6.60944	3.22430	2.04988	.53594
2.42	.52100	6.66579	3.23665	2.05947	.53175
2.43	.51996	6.72238	3.24894	2.06910	.52758
2.44	.51894	6.77919	3.26117	2.07876	.52344
2.45	.51792	6.83624	3.27335	2.08846	.51931
2.46	.51691	6.89353	3.28546	2.09819	.51521
2.47	.51592	6.95104	3.29751	2.10796	.51113
2.48	.51493	7.00879	3.30951	2.11777	.50707
2.49	.51395	7.06677	3.32145	2.12762	.50303
2.50	.51299	7.12499	3.33333	2.13750	.49902
2.51	.51203	7.18344	3.34515	2.14742	.49502
2.52	.51109	7.24212	3.35692	2.15737	.49105
2.53	.51015	7.30104	3.36863	2.16736	.48711
2.54	.50923	7.36019	3.38028	2.17739	.48318
2.55	.50831	7.41957	3.39187	2.18746	.47928
2.56	.50741	7.47919	3.40341	2.19756	.47540
2.57	.50651	7.53904	3.41488	2.20770	.47155
2.58	.50562	7.59912	3.42631	2.21788	.46772
2.59	.50474	7.65944	3.43767	2.22809	.46391
2.60	.50387	7.71999	3.44898	2.23834	.46012
2.61	.50301	7.78077	3.46023	2.24863	.45636
2.62	.50216	7.84179	3.47143	2.25895	.45263
2.63	.50131	7.90304	3.48257	2.26932	.44891
2.64	.50048	7.96452	3.49365	2.27971	.44522
2.65	.49965	8.02624	3.50468	2.29015	.44156
2.66	.49883	8.08819	3.51565	2.30062	.43792
2.67	.49802	8.15037	3.52657	2.31114	.43430
2.68	.49722	8.21279	3.53743	2.32168	.43071
2.69	.49642	8.27544	3.54824	2.33227	.42714
2.70	.49563	8.33832	3.55899	2.34289	.42359
2.71	.49485	8.40144	3.56969	2.35355	.42007
2.72	.49408	8.46479	3.58033	2.36425	.41657
2.73	.49332	8.52837	3.59092	2.37498	.41310
2.74	.49256	8.59219	3.60146	2.38575	.40965
2.75	.49181	8.65624	3.61194	2.39656	.40623
2.76	.49107	8.72052	3.62237	2.40741	.40283
2.77	.49033	8.78504	3.63274	2.41829	.39945
2.78	.48960	8.84979	3.64306	2.42922	.39610
2.79	.48888	8.91477	3.65333	2.44017	.39277
2.80	.48817	8.97999	3.66355	2.45117	.38946
2.81	.48746	9.04544	3.67371	2.46221	.38618
2.82	.48676	9.11112	3.68383	2.47328	.38293
2.83	.48606	9.17704	3.69388	2.48439	.37970
2.84	.48538	9.24319	3.70389	2.49553	.37649
2.85	.48469	9.30957	3.71385	2.50672	.37330
2.86	.48402	9.37619	3.72375	2.51794	.37014
2.87	.48335	9.44304	3.73361	2.52920	.36700
2.88	.48269	9.51012	3.74341	2.54050	.36389
2.89	.48203	9.57744	3.75316	2.55183	.36080
2.90	.48138	9.64499	3.76286	2.56321	.35773
2.91	.48074	9.71277	3.77251	2.57462	.35469
2.92	.48010	9.78079	3.78211	2.58606	.35167
2.93	.47946	9.84904	3.79166	2.59755	.34867
2.94	.47884	9.91752	3.80117	2.60907	.34570
2.95	.47821	9.98624	3.81062	2.62064	.34275
2.96	.47760	10.05519	3.82002	2.63223	.33982
2.97	.47699	10.12437	3.82937	2.64387	.33692
2.98	.47638	10.19379	3.83868	2.65555	.33404
2.99	.47578	10.26344	3.84793	2.66726	.33118
3.00	.47519	10.33333	3.85714	2.67901	.32834
3.50	.45115	14.12500	4.26087	3.31505	.21295
4.00	.43496	18.50000	4.57143	4.04688	.13876
4.50	.42355	23.45833	4.81188	4.87509	.09170
5.00	.41523	29.00000	5.00000	5.80000	.06172
6.00	.40416	41.83333	5.26829	7.94059	.02965
7.00	.39736	57.00000	5.44444	10.46939	.01535
8.00	.39289	74.50000	5.56522	13.38672	.00849
9.00	.38980	94.33334	5.65116	16.69273	.00496
10.00	.38758	116.50000	5.71429	20.38750	.00304
∞	.37796	∞	6.00000	1	0

xxxiv | xxxv | xxxvi | xxxvii | xxxviii

Anhang B1

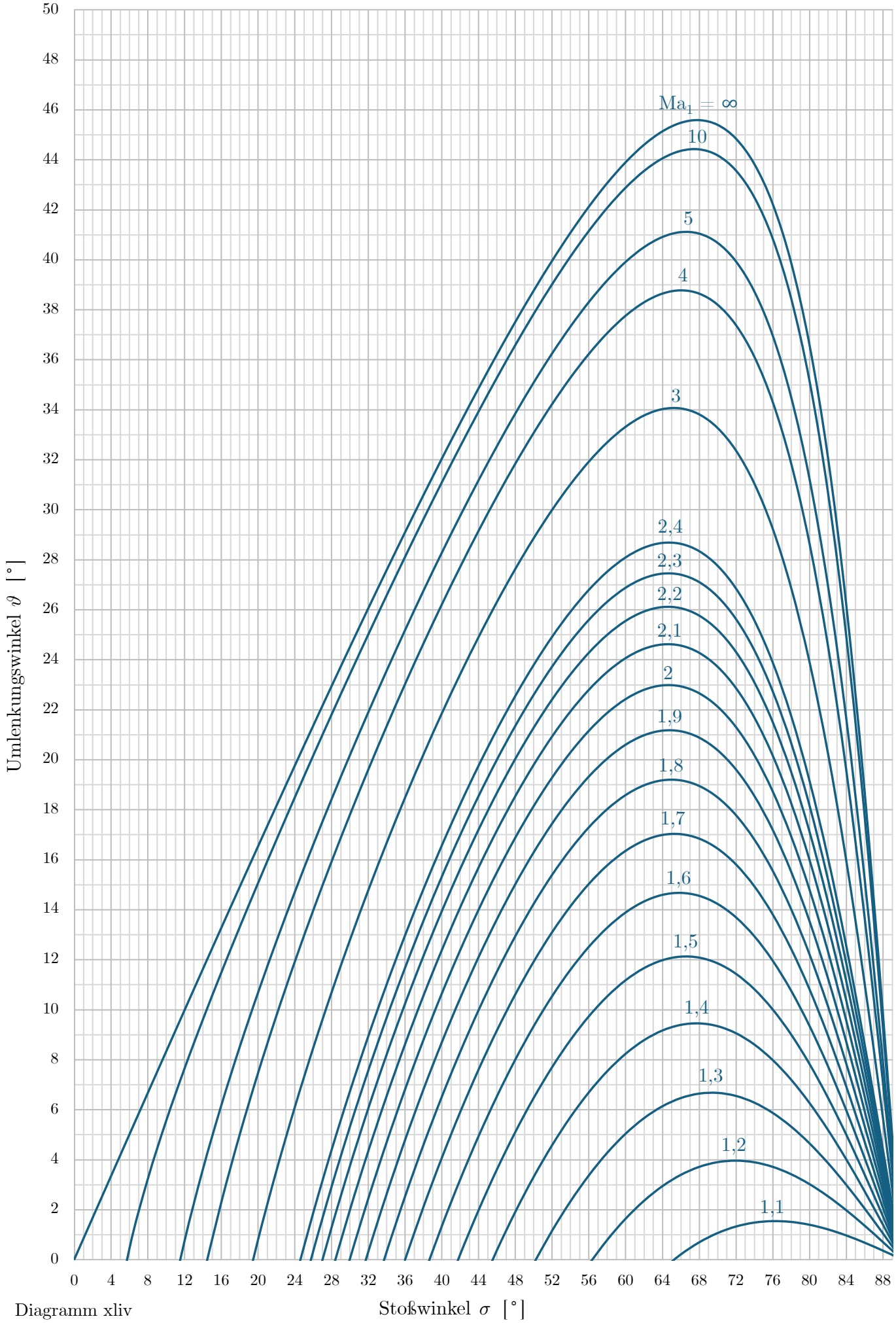

Diagramm xliv

Anhang B2

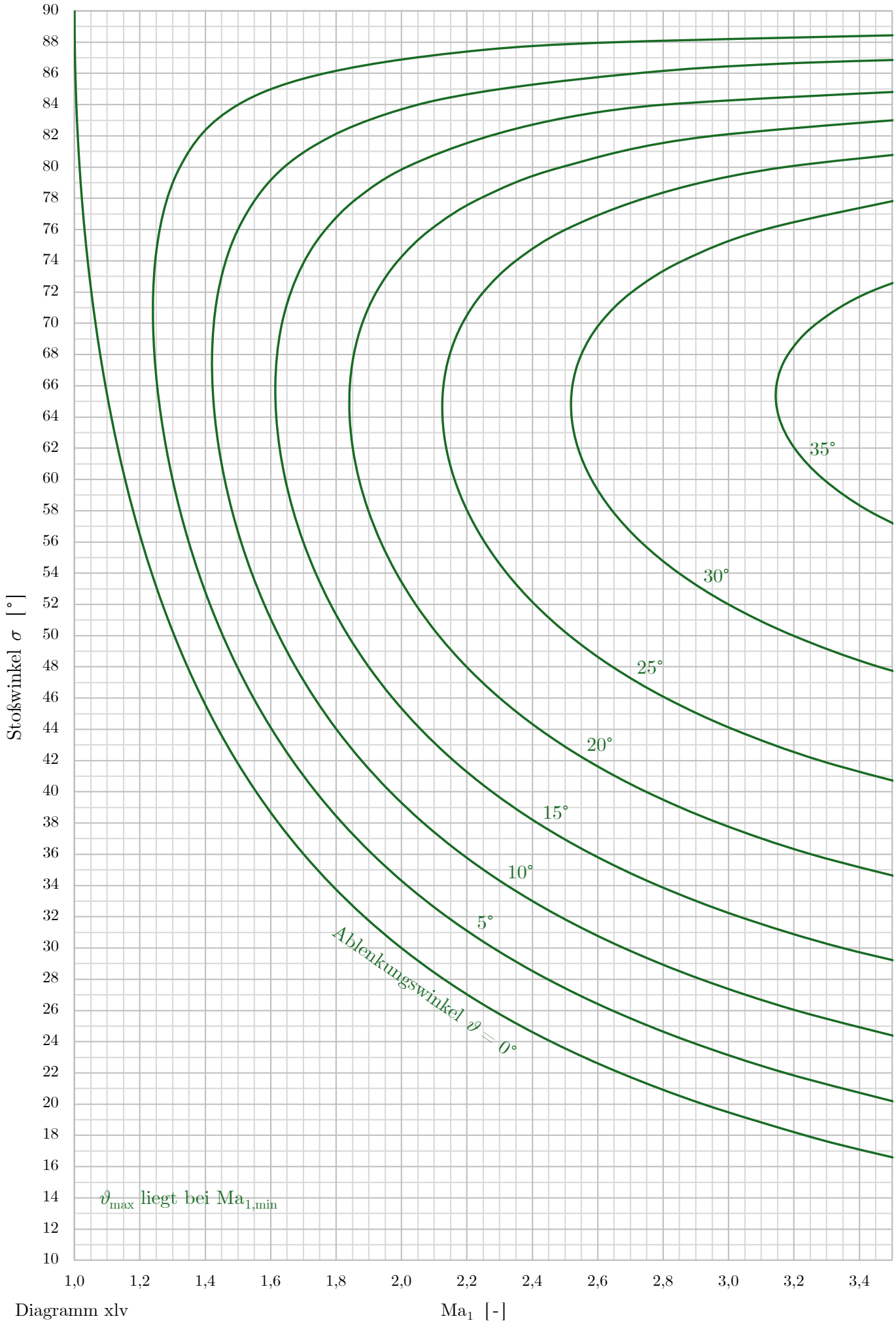

Diagramm xlv

Anhang B3

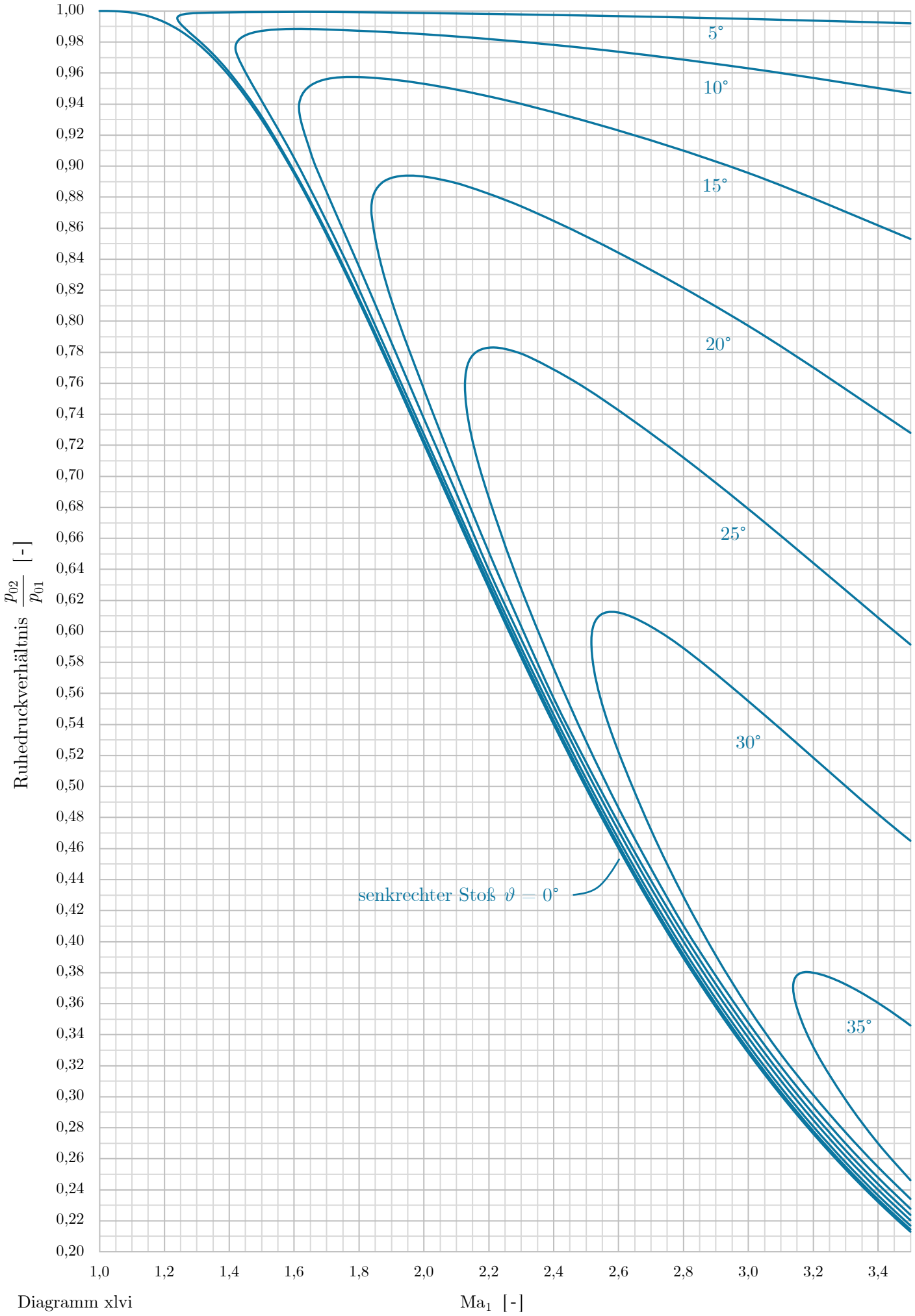

Diagramm xlvi

Anhang B4

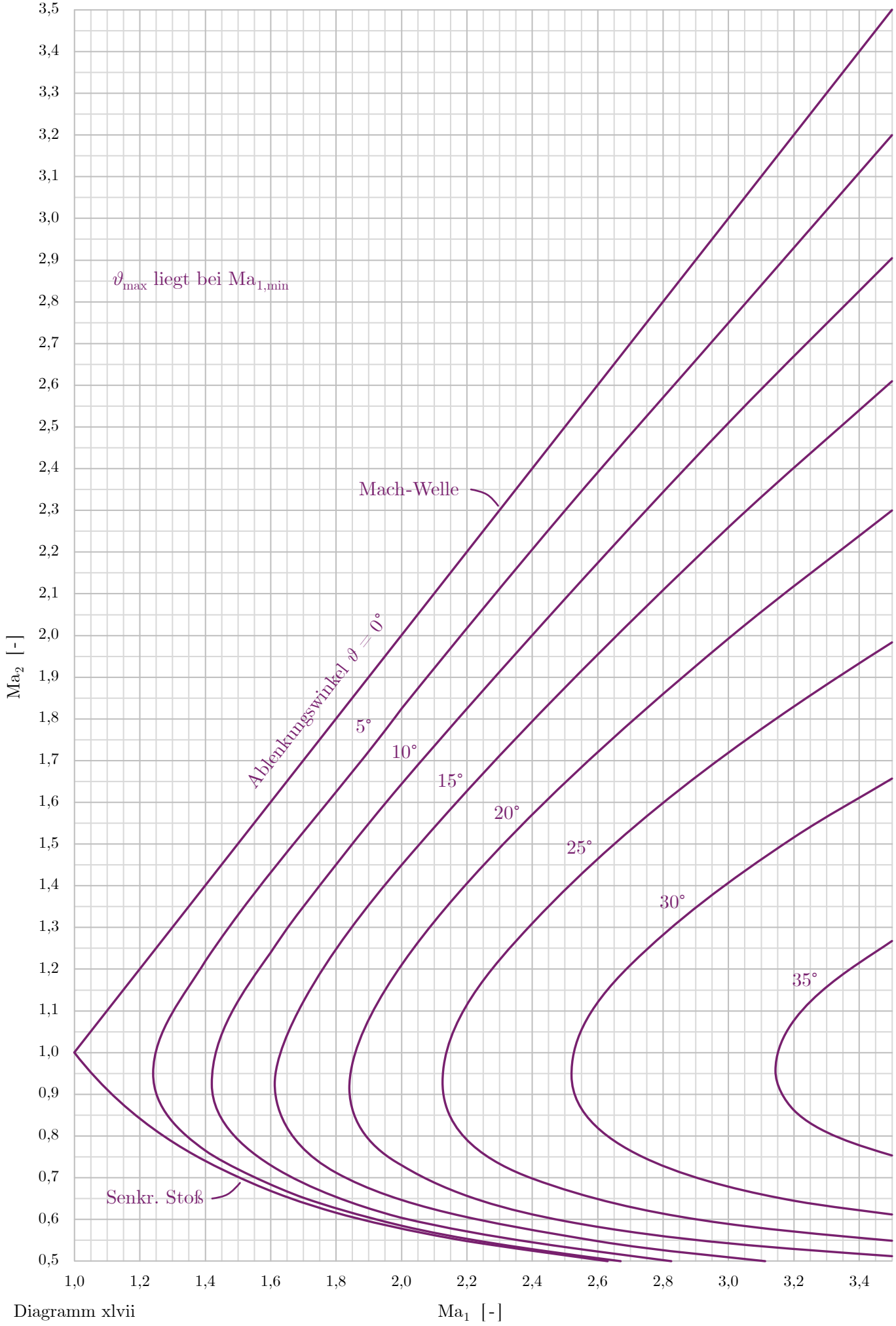

Diagramm xlvii

Anhang B5

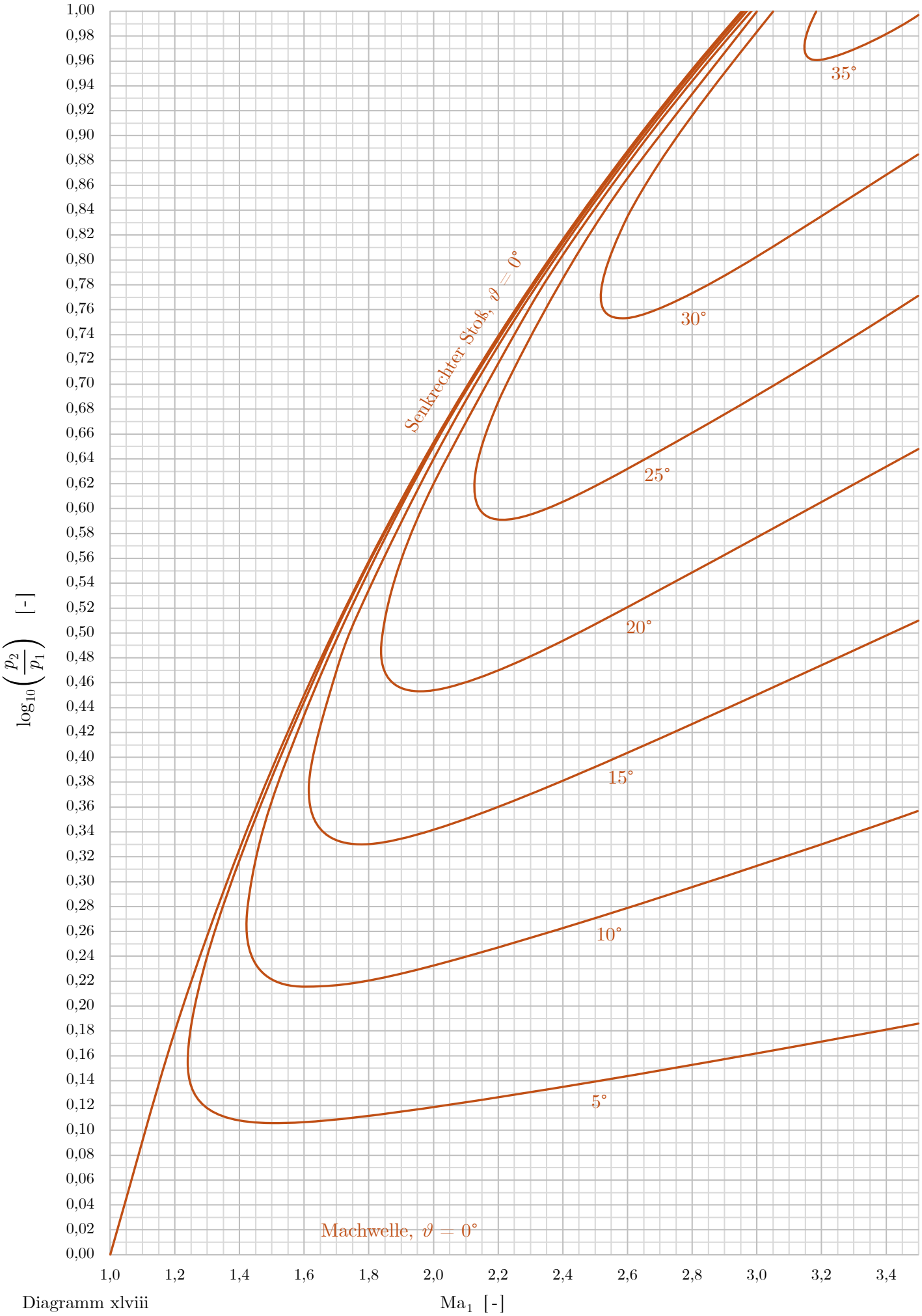

Diagramm xlviii